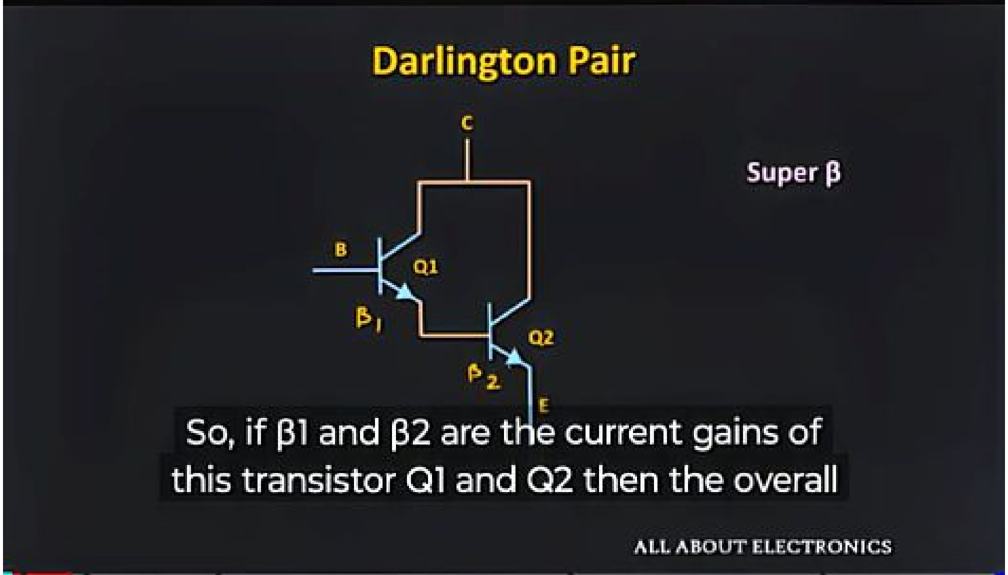
11:45 0 KB/s 5G+ 52%



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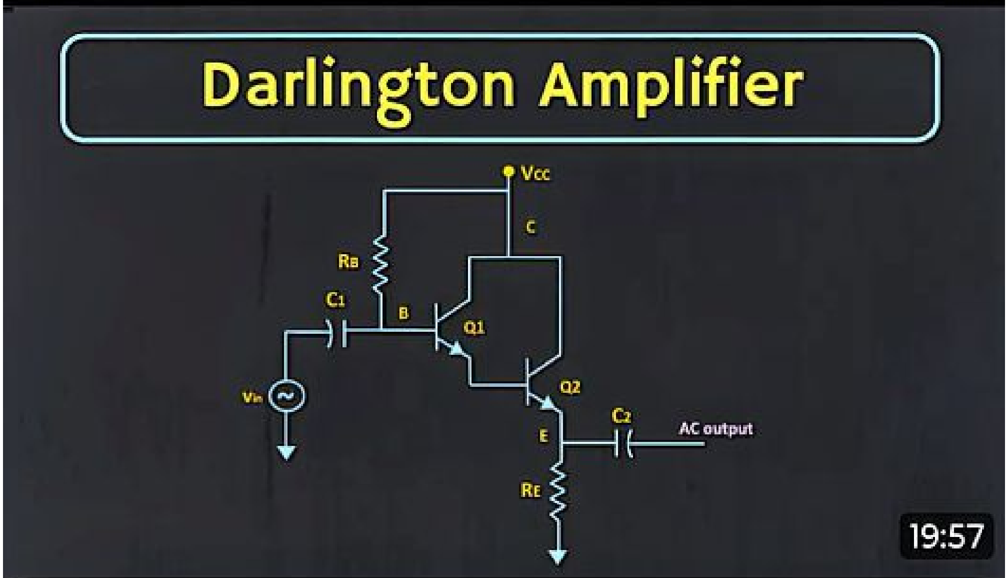

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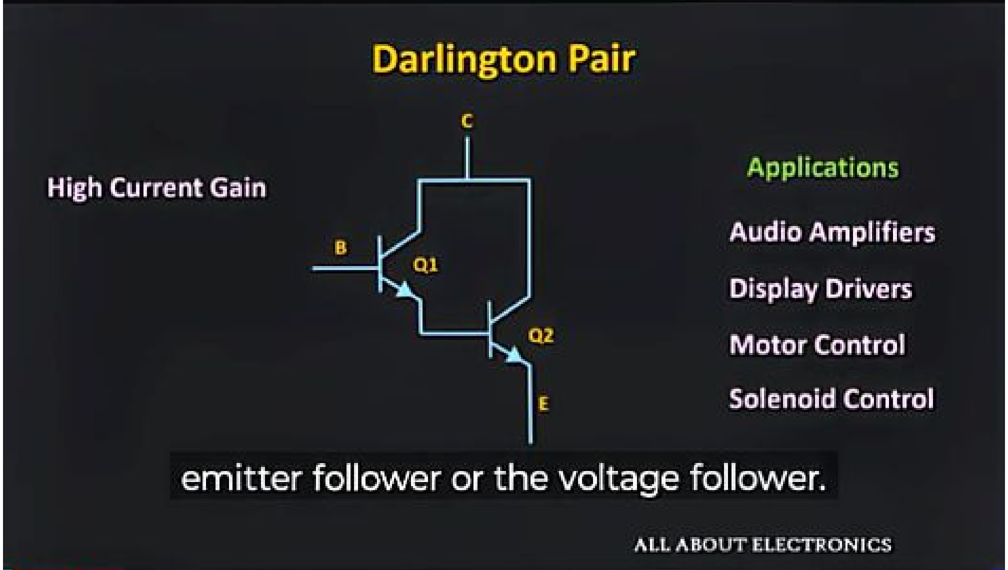

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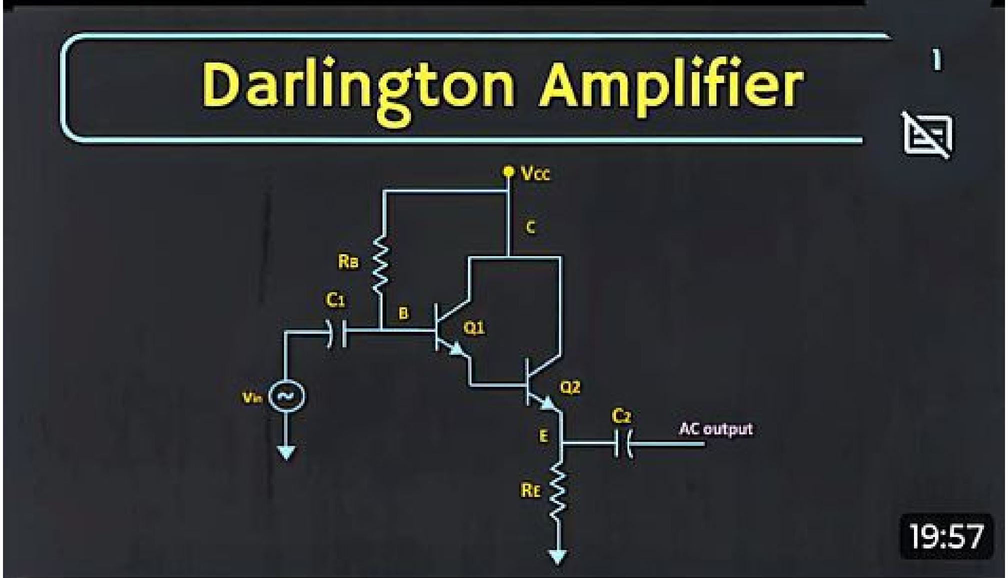

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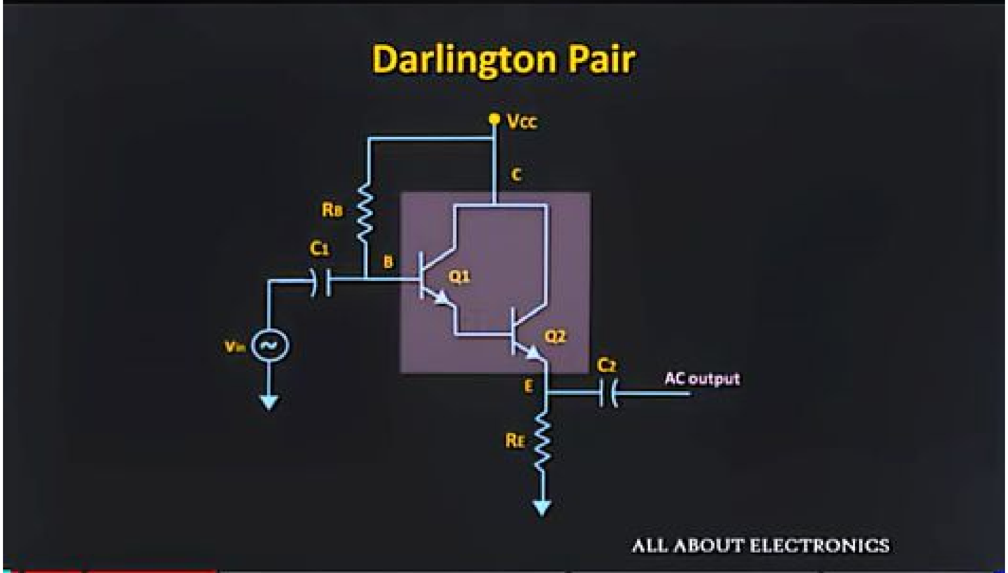

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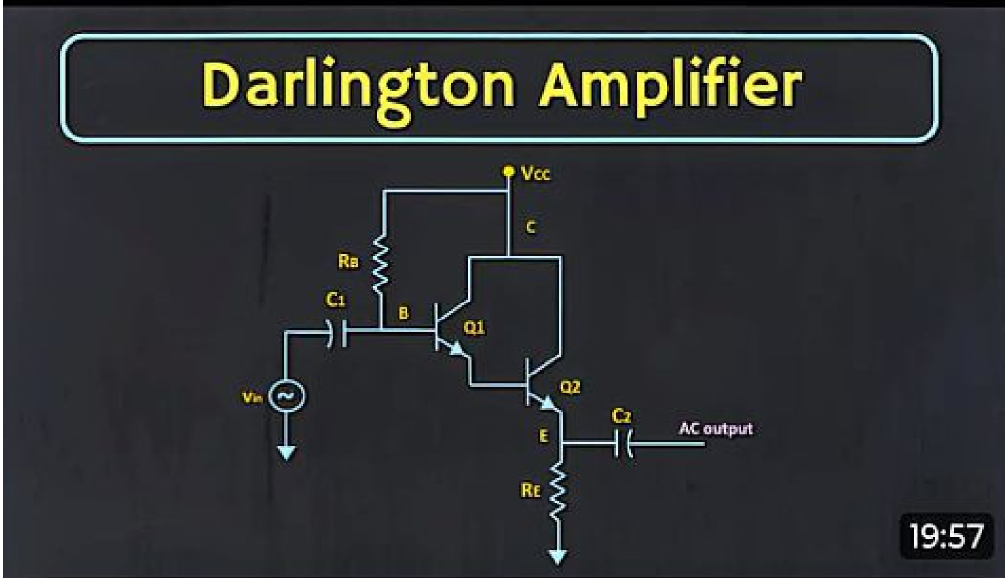

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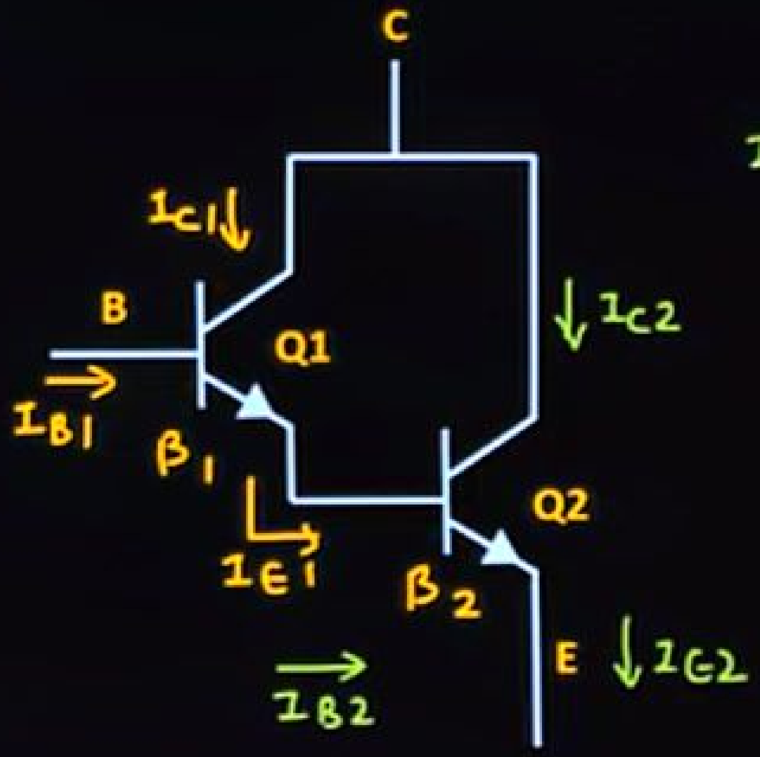
Darlington Pair



$$I_{B2} = I_{E1}$$

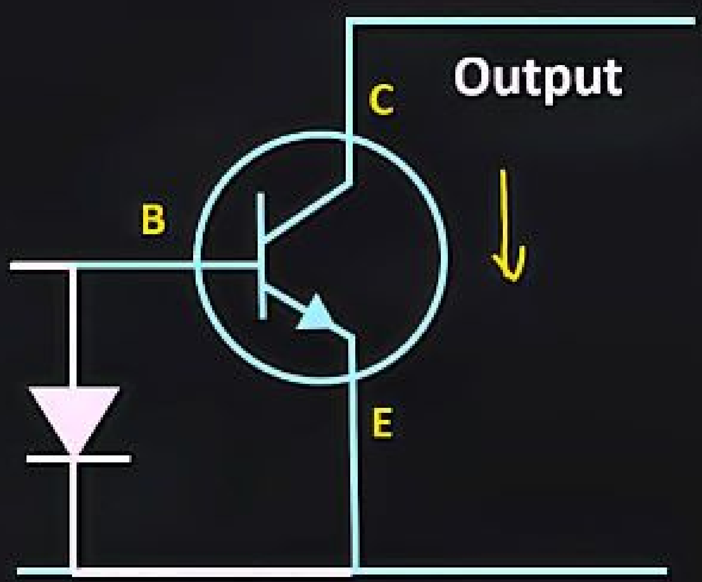
$$I_{B2} = I_{E1} = (\beta_1 + 1) I_{B1}$$

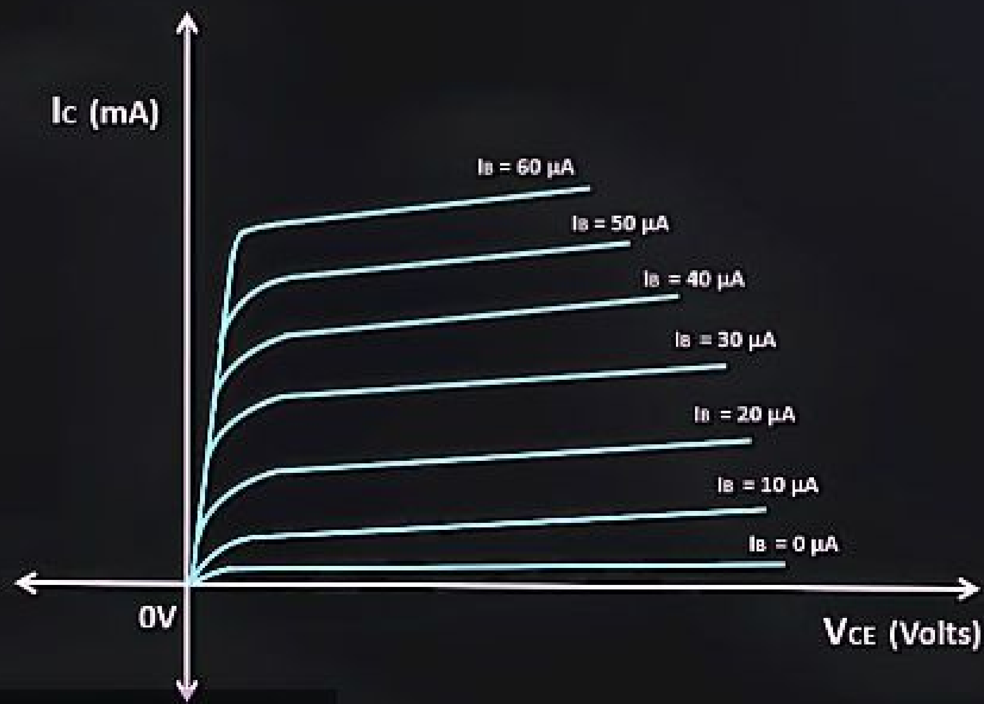
$$I_{C2} = \beta_2 I_{B2}$$

$$= \beta_2 (\beta_1 + 1) I_{B1}$$

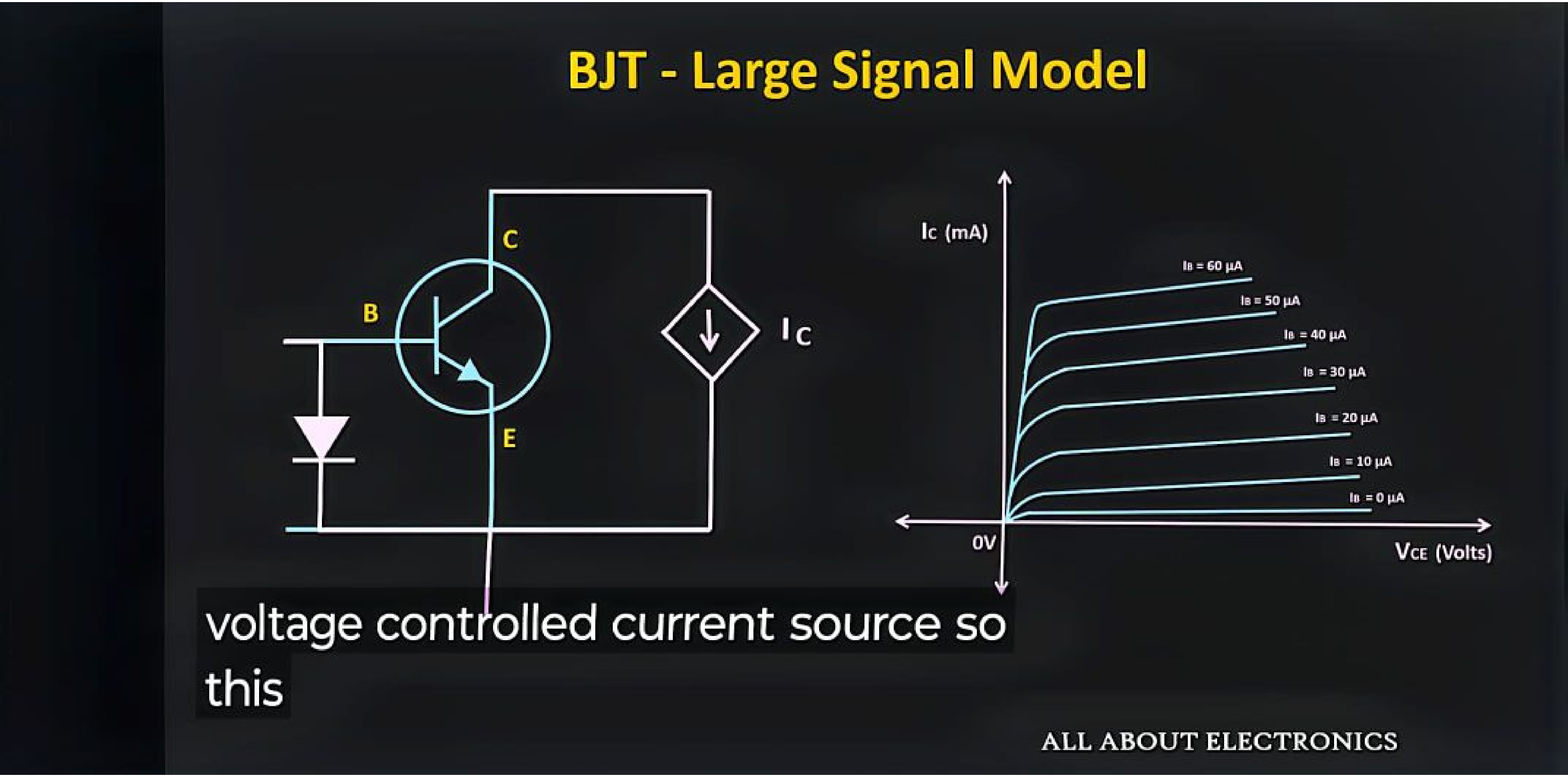
$$I_{C1} = \beta_1 I_{B1}$$

BJT - Large Signal Model





assumed to be a constant for the given value of VB or we


BJT - Large Signal Model
C
B
E
I C
Ic (mA)
IB = 60 µA
IB = 50 µA
IB = 40 µA
IB = 30 µA
IB = 20 µA
IB = 10 µA
IB = 0 µA
0V
VCE (Volts)
voltage controlled current source so
this
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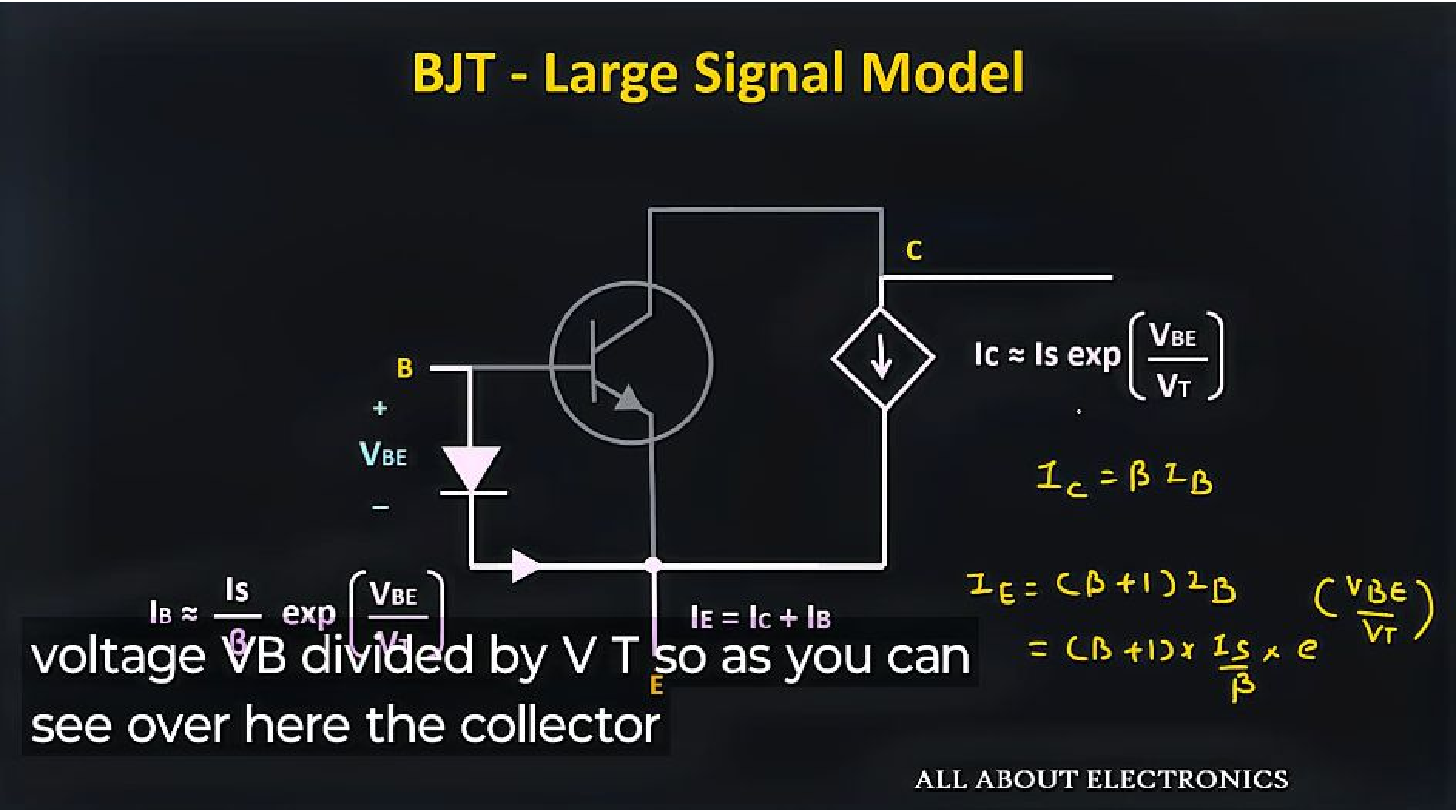
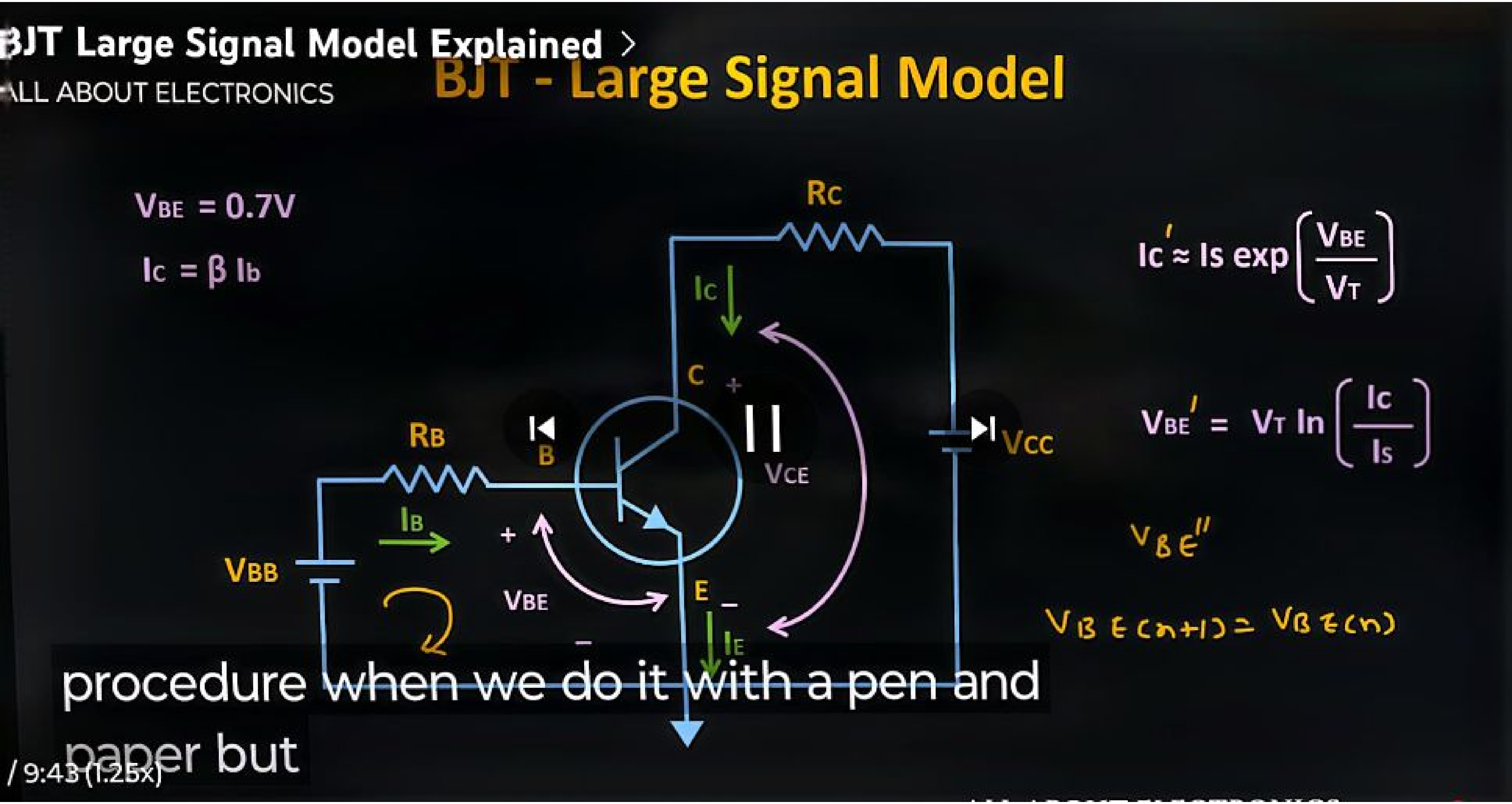

BJT - Large Signal Model
C
B
+
VBE
−
E
Ic ≈ Is exp (VBE / VT)
IB ≈ Is/β exp (VBE / VT)
IE = IC + IB
IC = β IB
IE = (β + 1) IB
= (β + 1) × Is/β × e^(VBE/VT)
voltage VB divided by V T so as you can
see over here the collector
ALL ABOUT ELECTRONICS

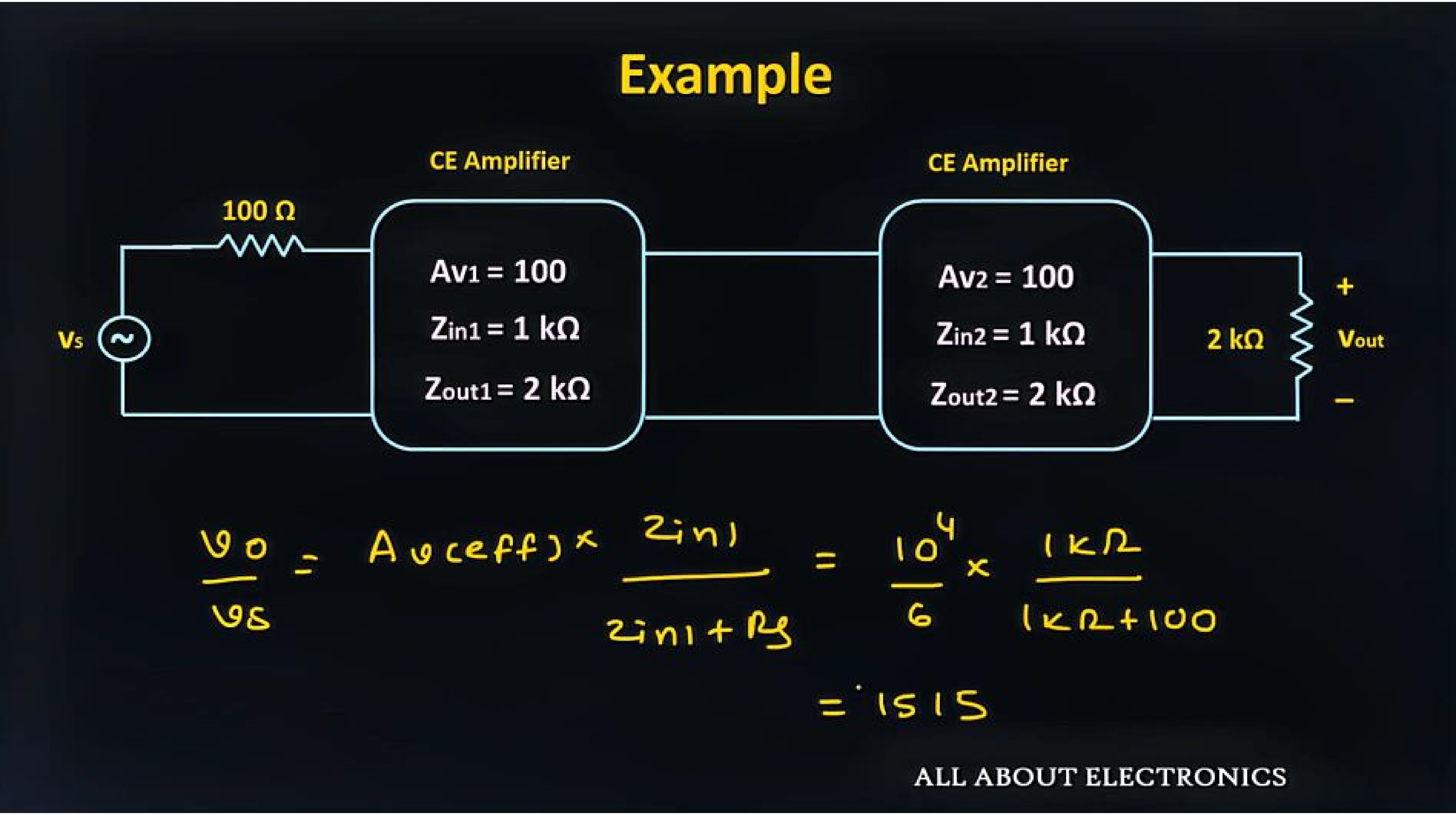


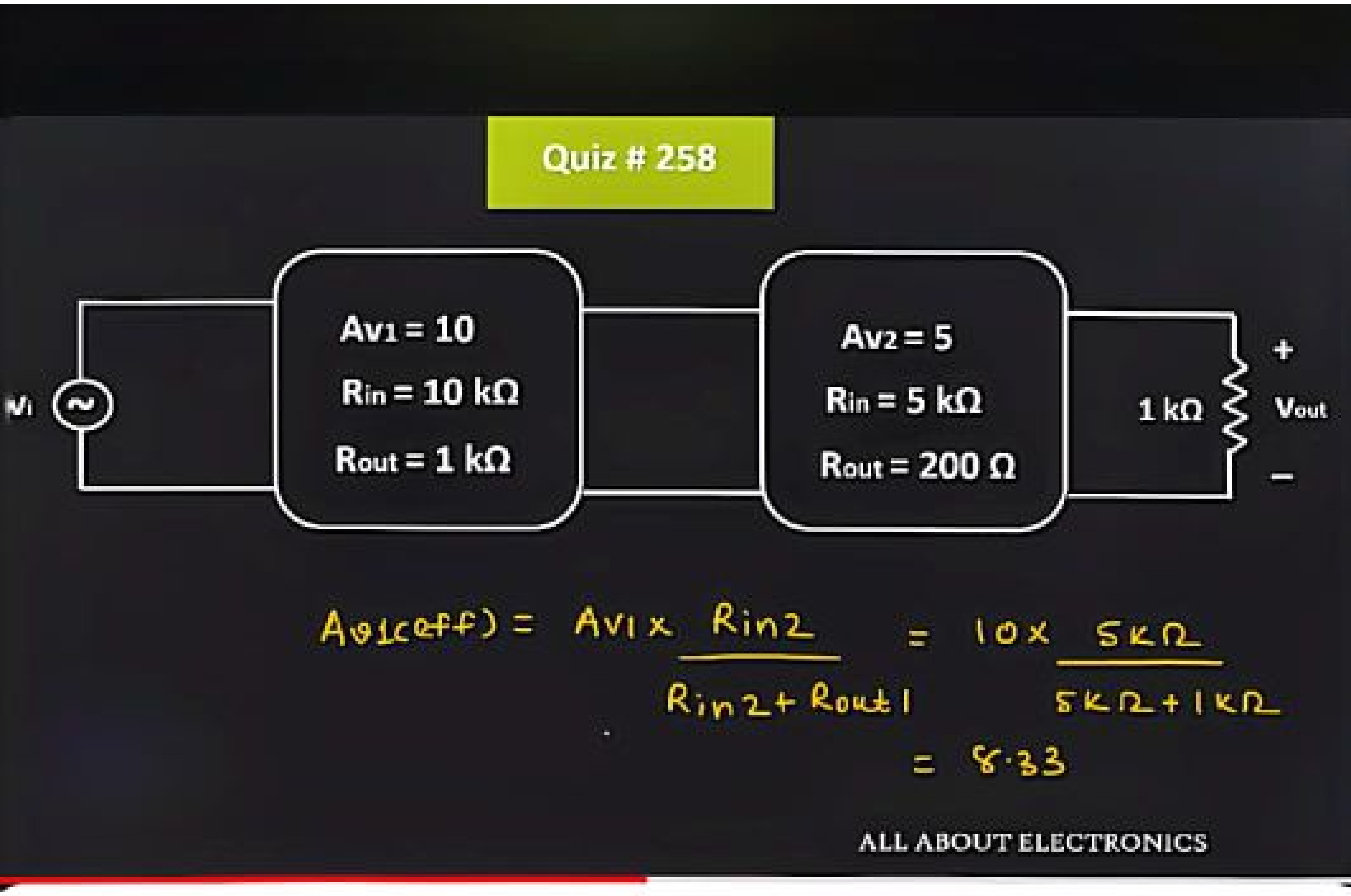
BJT Large Signal Model Explained
ALL ABOUT ELECTRONICS
BJT - Large Signal Model
$V_{BE} = 0.7V$
$I_C = \beta I_b$
$I_C' \approx I_S \exp\left(\frac{V_{BE}}{V_T}\right)$
$V_{BE}' = V_T \ln\left(\frac{I_C}{I_S}\right)$
V_{BE}''
$V_{BE(n+1)} = V_{BE(n)}$
R_C
R_B
I_C
C
B
E
V_{CE}
V_{CC}
V_{BB}
I_B
V_{BE}
I_E
procedure when we do it with a pen and
paper but
/ 9:43 (1.25x)

BJT- Small Signal Model

$\Delta I_B = \frac{\Delta I_C}{\beta}$

$I_C = \beta I_B$


Example
CE Amplifier
CE Amplifier
100 Ω
Vs
Av1 = 100
Zin1 = 1 kΩ
Zout1 = 2 kΩ
Av2 = 100
Zin2 = 1 kΩ
Zout2 = 2 kΩ
2 kΩ
+
Vout
–
$\frac{v_o}{v_s} = A_{v(eff)} \times \frac{Z_{in1}}{Z_{in1} + R_s} = \frac{10^4}{6} \times \frac{1k\Omega}{1k\Omega + 100}$
$= 1515$
ALL ABOUT ELECTRONICS



Multistage Amplifier | Quiz # 258

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Comments 1

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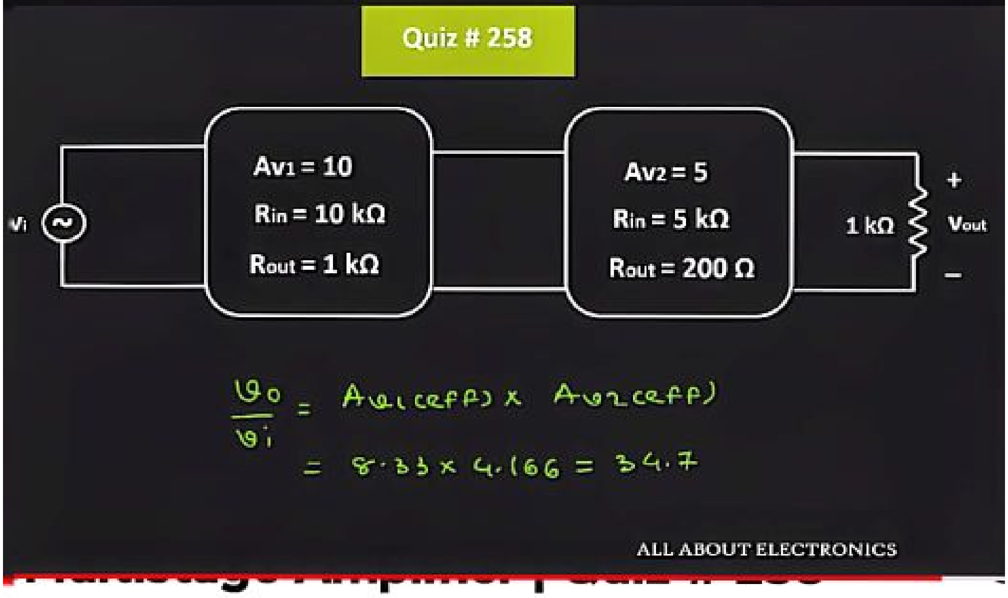

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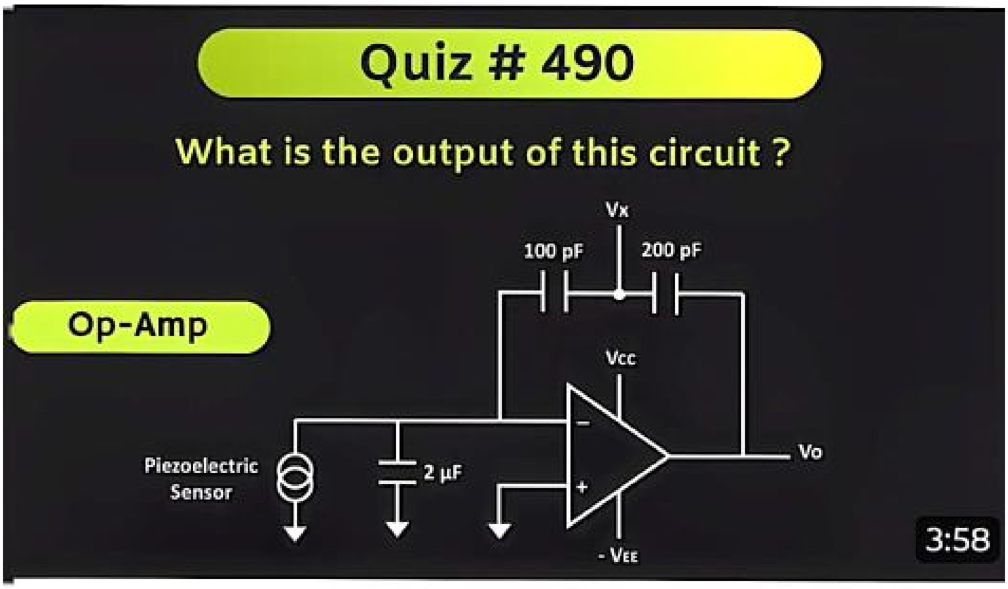
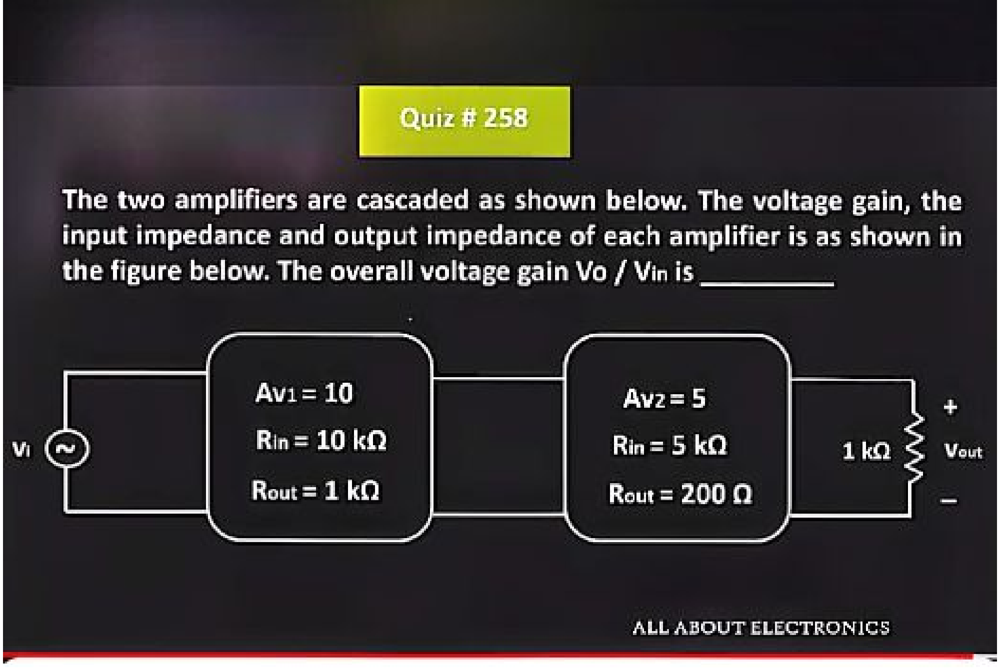


Operational Amplifier (Op-Amp) Solved Problem | Quiz # 490

Next:Darlington Pair Explained | The Darlington...
Analog Electronics • 121/328



Multistage Amplifier | Quiz # 258

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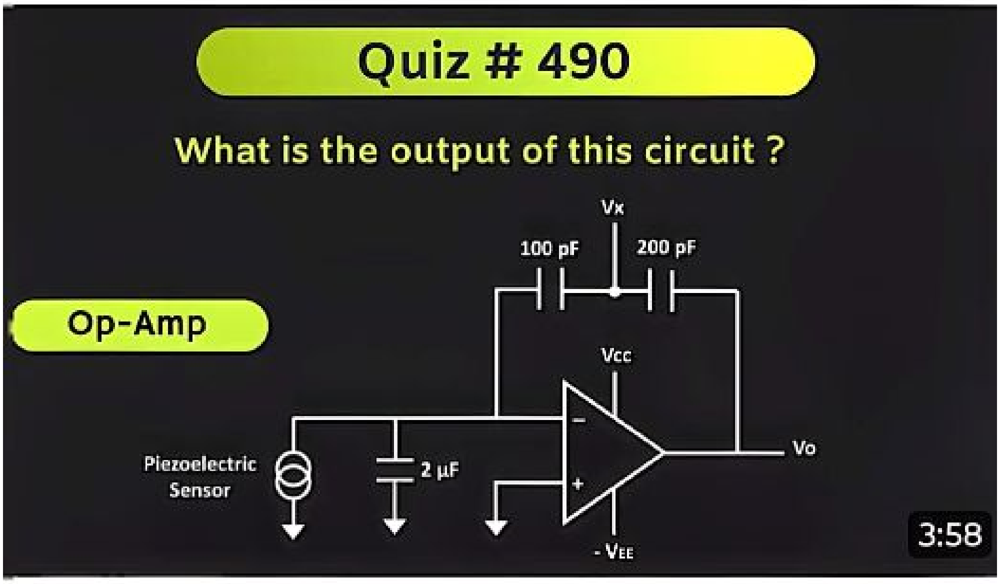

Operational Amplifier (Op-Amp) Solved

Next:Darlington Pair Explained | The Darlington...

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MOSFET Transconductance and MOSFET Small Signal Model Explained

1.6 lakh views 3 yr ago #MOSFET ...more

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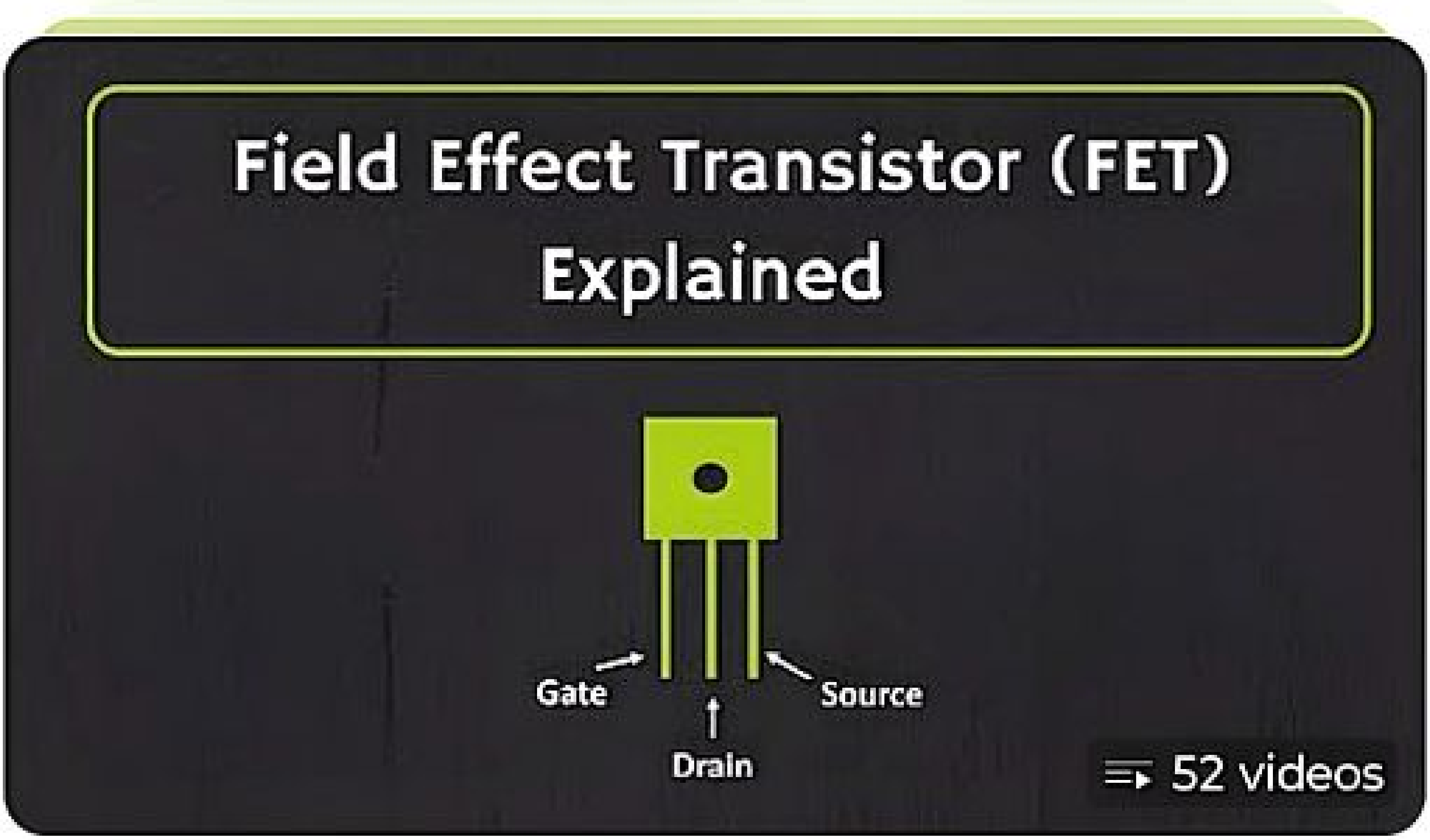
2.3K • 64 Share Thanks

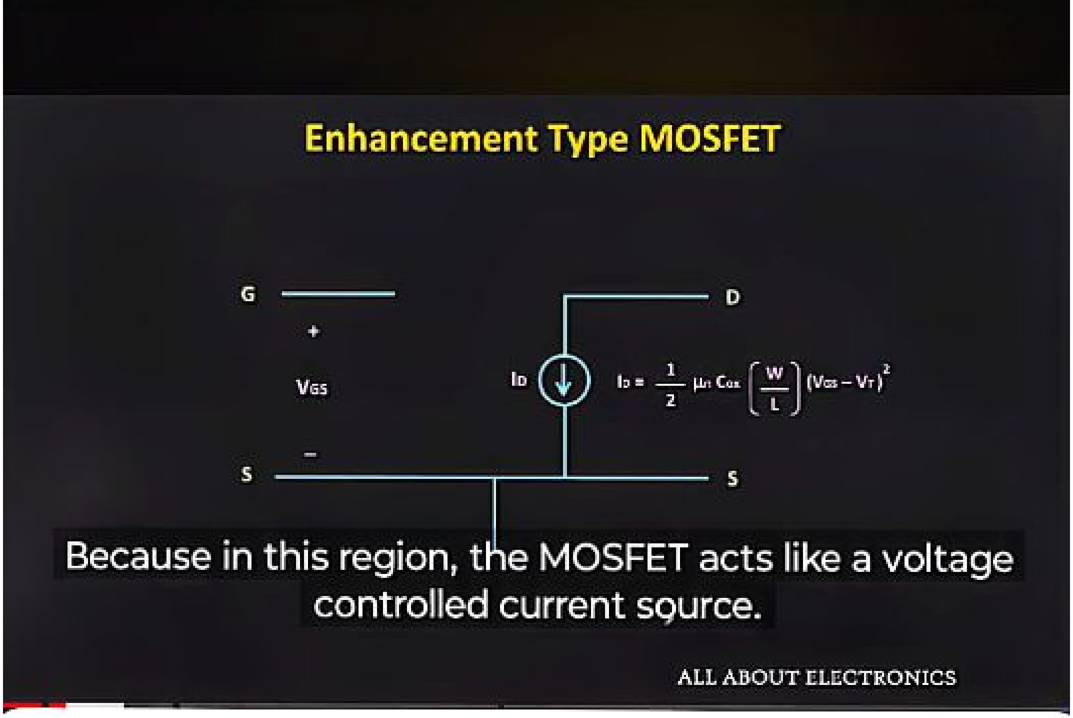
Comments 63

V Add a comment...



Next:MOSFET Transconductance | Quiz # 309

Subtitles/CC turned on (English)



MOSFET Transconductance and MOSFET Small Signal Model Explained

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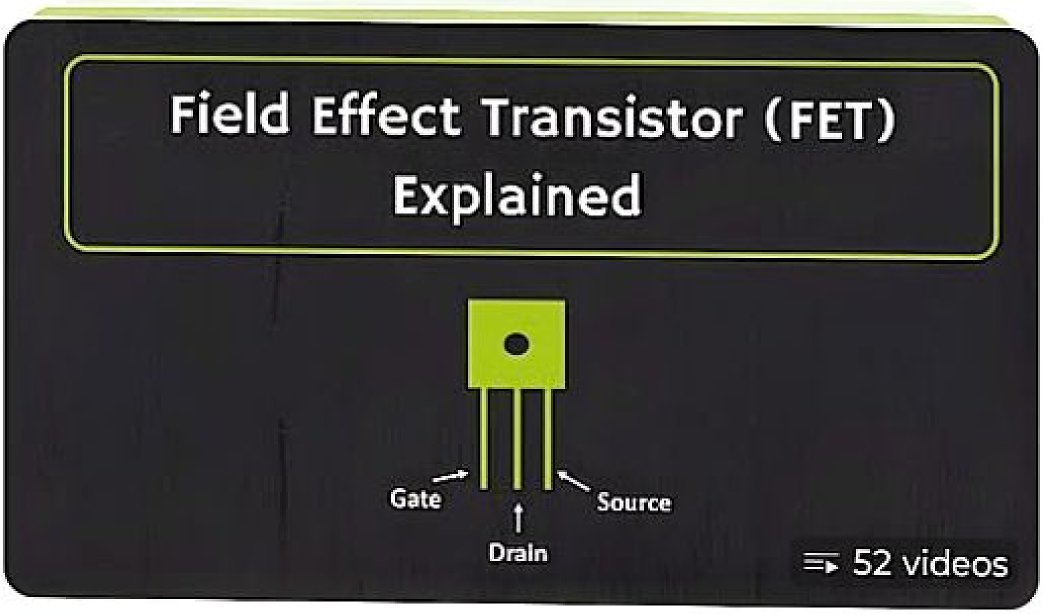
Comments 63

V Add a comment...



Next:MOSFET Transconductance | Quiz # 309

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MOSFET Transconductance and MOSFET Small Signal Model Explained

1.6 lakh views 3 yr ago #MOSFET ...more

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Comments 63

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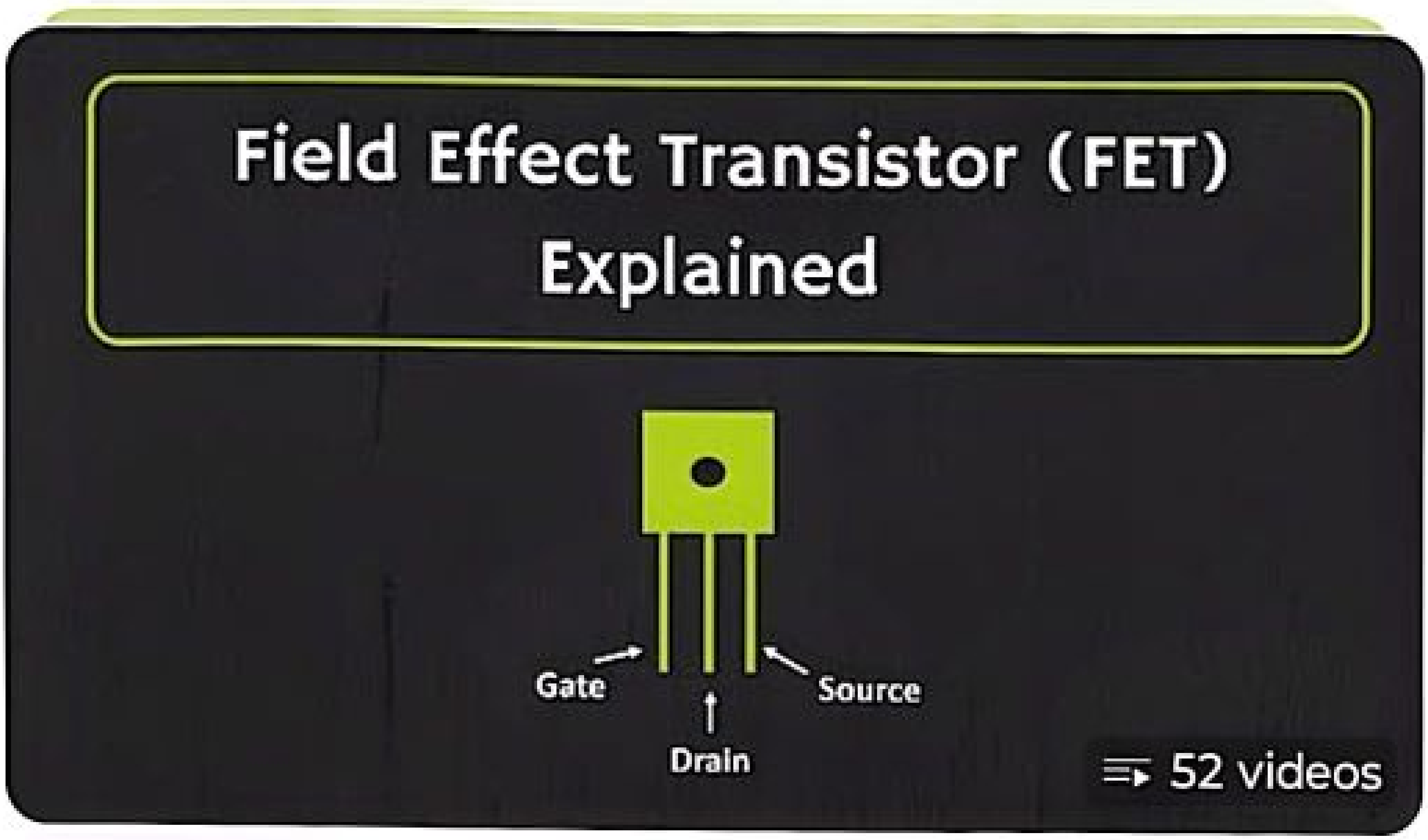

Next:MOSFET Transconductance | Quiz # 309

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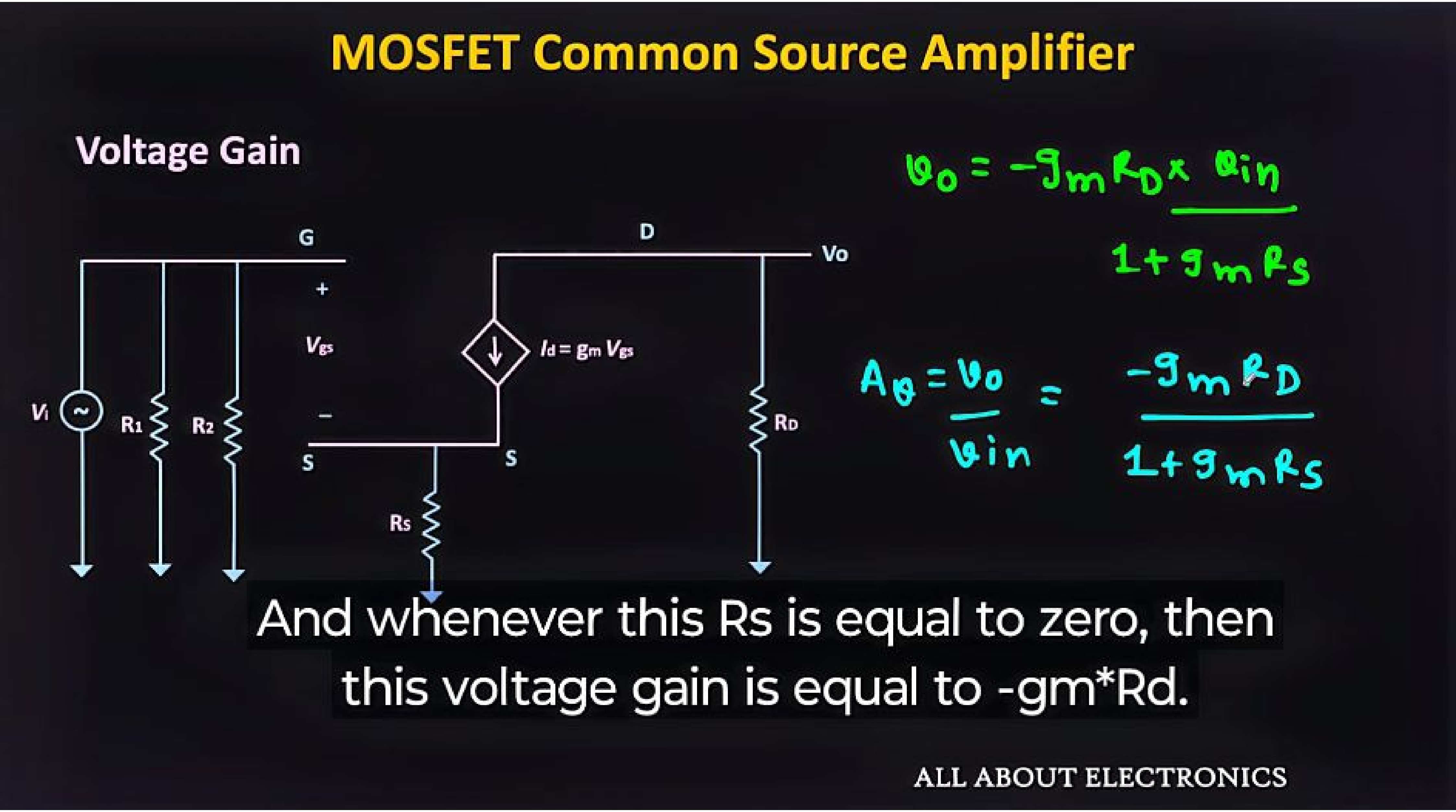
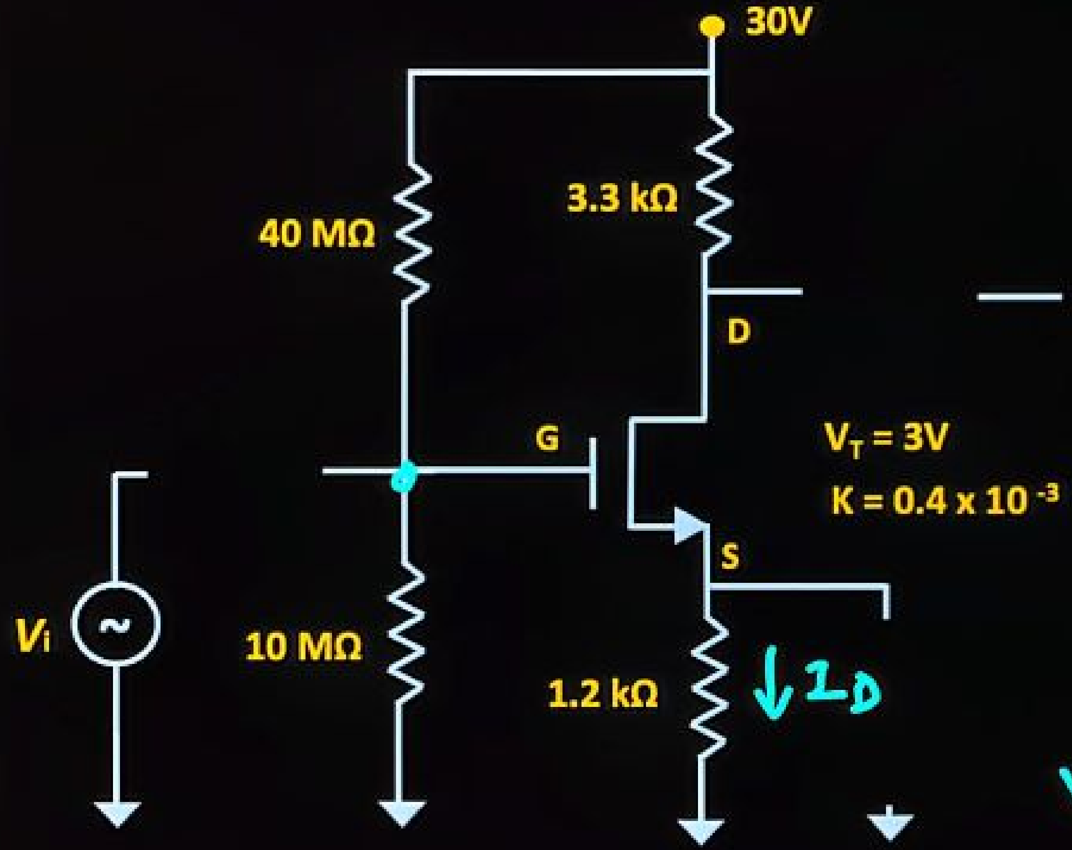

MOSFET Common Source Amplifier
Voltage Gain
G
D
Vo
+
Vgs
Id = gm Vgs
–
S
S
Vi
R1
R2
Rs
RD
$v_o = -g_m R_D \times \frac{v_{in}}{1+g_m R_S}$
$A_v = \frac{v_o}{v_{in}} = \frac{-g_m R_D}{1+g_m R_S}$
And whenever this Rs is equal to zero, then
this voltage gain is equal to -gm*Rd.
ALL ABOUT ELECTRONICS

Example



DC Analysis

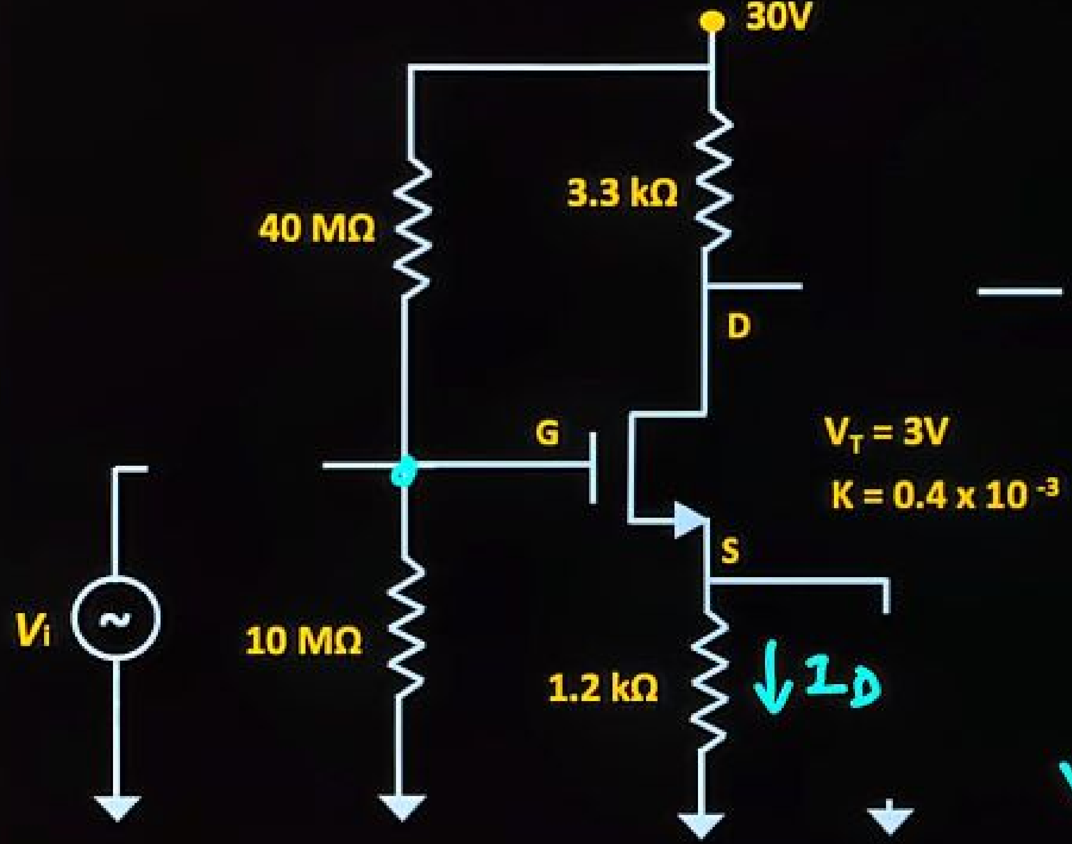
$$V_g = \frac{10M\Omega}{10M\Omega + 40M\Omega} \times 30$$

$$= 6V$$

$$V_S = R_S I_D$$

$$V_{GS} =$$

Example



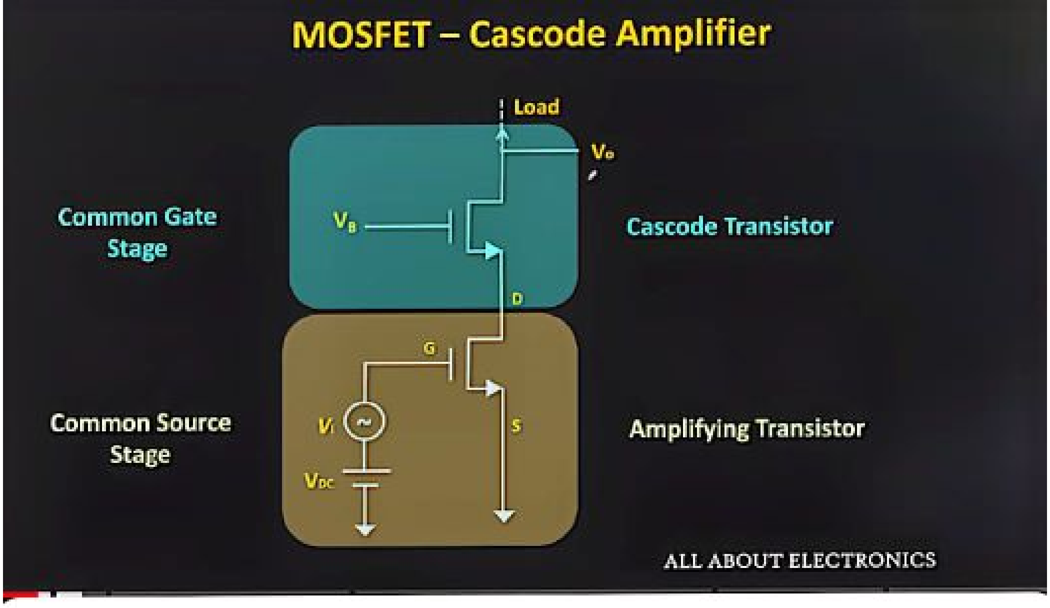
DC Analysis

$$V_g = \frac{10M\Omega}{10M\Omega + 40M\Omega} \times 30$$

$$= 6V$$

$$V_S = R_S I_D$$

$$V_{GS} = 6V - I_D R_S$$



Cascode Amplifier using MOSFET Explained (Cascode Amplifier with Cas...

88K views 3 yr ago #MOSFET ...more

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1K 28 Share Download

Comments 31

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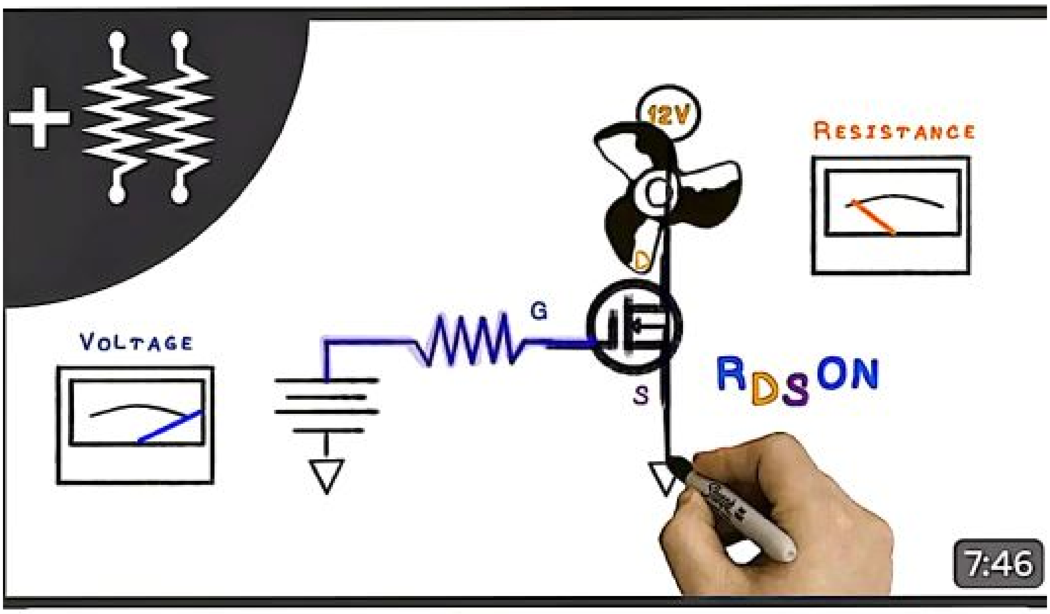


Next:MOSFET- Cascode Amplifier | Quiz # 320
Analog Electronics • 168/328

AddOhms • 37 lakh views • 10 years ago

Characteristics Of Logic families / Important Parameters of Digital IC

① **Speed** : Determined by time betⁿ the application of input and change in out put.

② **Fan out** : The number of gates that can be driven by gate out put.

③ **Fan in** : The maxᵐ number of inputs that can be applied to a logic gate.

④ **Power dissipation** : Power used by per gate.

⑤ **Propagation Dealy** : Time taken by t[illegible]l to reach to out put from input

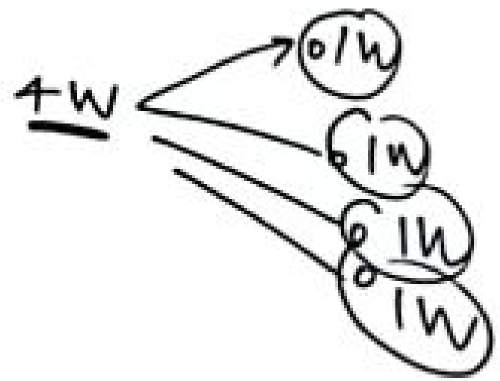
[illegible] **Margin** : Limit of Noise voltage that [illegible] present without disturbing [illegible] operation of circuit.

⑦ **Figure of time** : It is the product of propagation dealy & Power dissipation.

⑧ **Logic Swing** : It is the difference between two output Voltage.

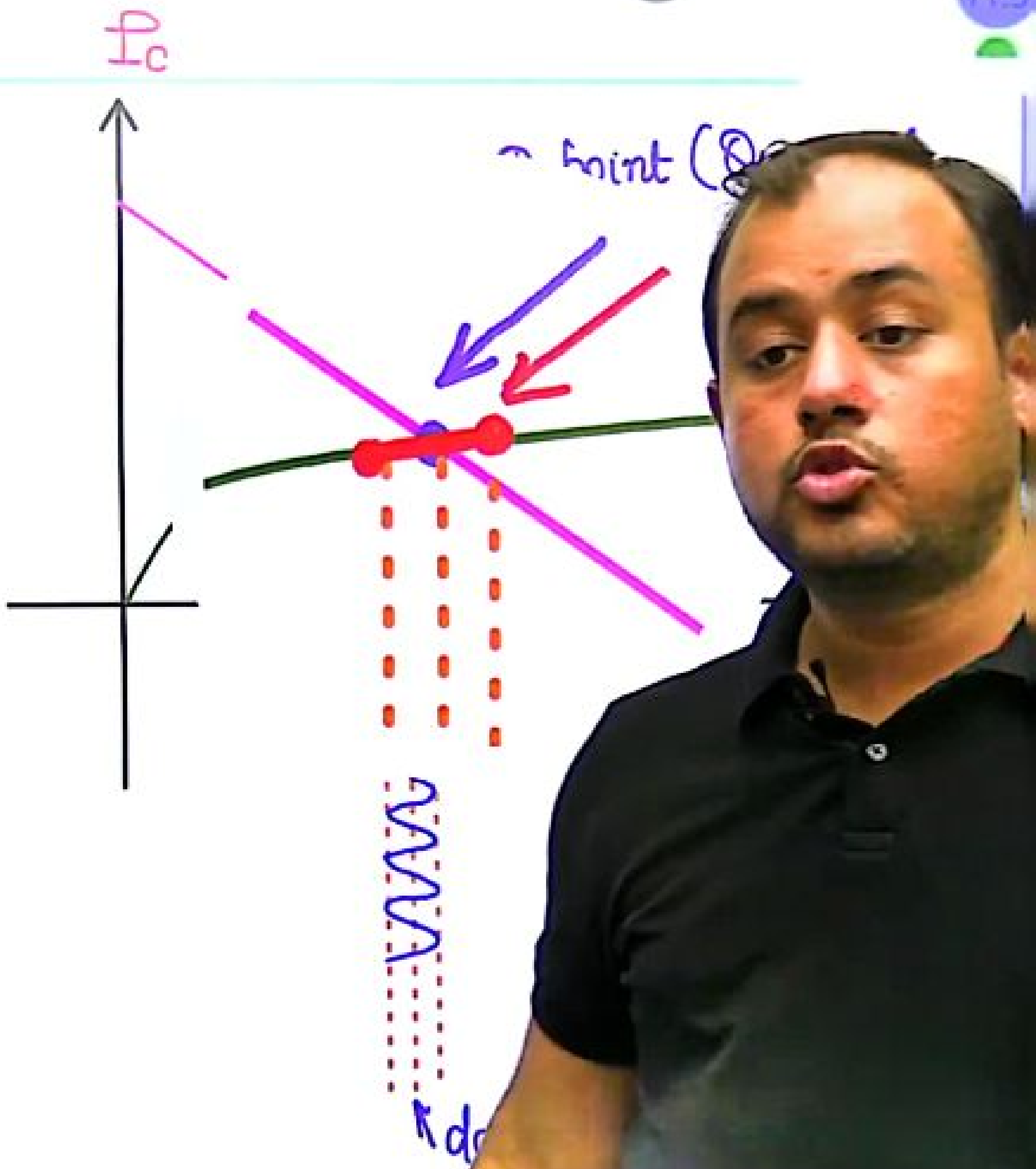
⑨ **Noise Immunity** : Maxᵐ noise that a circuit can with stand without changing the out put.

⑩ **Breadth** : The number of various functions available in a logic family.



Small Signal Analysis

- we superimpose a small ac signal on a large dc signal due to which operating point fluctuates a little about Q-point.
- Since fluctuation is small we can linearize the char. in such small region & be able to use Superpos^n Theorem.



How to decide Q-Point?

- If transistor is to be used as a switch, then Q-point must lie in cut off or saturation region.

  Saturation : ON switch

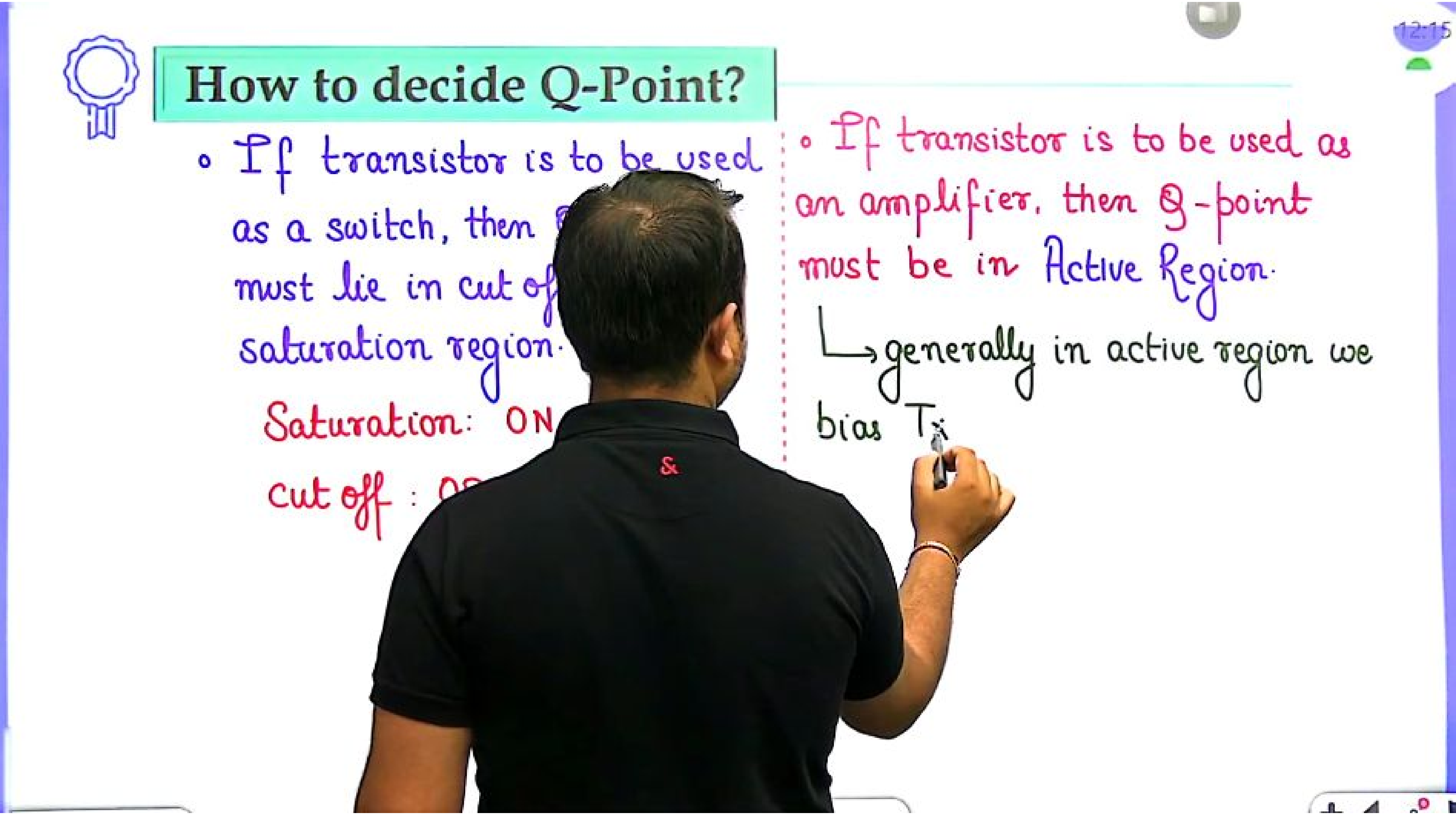
  cut off : OFF switch

- If transistor is to be used as an amplifier, then Q-point must b... ...tive Region.

  ↳ ... active region we bi... ...id of load line

  ↳ equidistant from ... sat & cut off

  ...ax^m gain


How to decide Q-Point?
• If transistor is to be used
as a switch, then
must lie in cut of
saturation region.
Saturation: ON
cut off :
• If transistor is to be used as
an amplifier, then Q-point
must be in Active Region.
generally in active region we
bias Tx
12:15

How to decide Q-Point?

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  Saturation : ON switch

  cut off : OFF switch

- If transistor is to be used as an amplifier, then Q-point must ... ctive Region.

  ↳ ... active region we ...

  b... mid of load line

  ↳ equidistant from

  sat & cut off

  ...ax^{m} gain ...

load line

$I_C = \frac{V_{CC} - V_{CE}}{\quad}$

at ... $V_{CC}/2R_C$

$\frac{V_{CC}}{2}$

Power dissipated

$= V_{CE} I_C$

$P = (V_{CC} - I_C R_C) I_C$

$\frac{dP}{dI_C} = V_{CC} - 2 I_C R_C = 0$

$I_C = V_{CC}/2R_C \leftarrow$ mid of load line

at mid of load line P_T is max^m

& $\eta =$ min^m

12:19

load line

$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

at mid, $I_C = V_{CC}/2R_C$

$$V_{CE} = \frac{V_{CC}}{2}$$

Power dissipated

$= V_{CE} I_C$

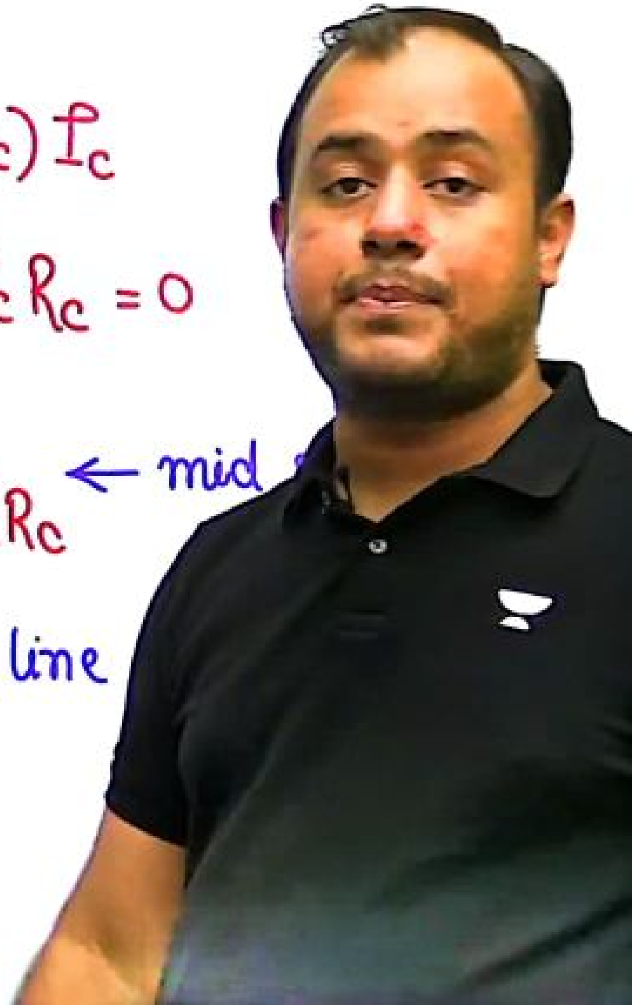
$P = (V_{CC} - I_C R_C) I_C$

$$\frac{dP}{dI_C} = V_{CC} - 2 I_C R_C = 0$$

$I_C = V_{CC}/2R_C$ ← mid

at mid of load line

& $\eta = \text{min}^{m}$



How to decide Q-Point?

- If transistor is to be used as a switch, then Q-point must lie in cut off or saturation region.

  Saturation : ON switch

  cut off : OFF switch

- If transistor is to be used as an amplifier, then Q-point must be in Active Region.

  ↳ generally in active region we bias Tx at mid of load line

    ↳ equidistant from sat & cut off

    ↳ max^m gain or amplification

Derivation of 'S'

- I_C is most sensitive to variation in I_{CBO}.

$$S = \frac{\partial I_C}{\partial I_{CBO}}$$

$$I_C = \beta I_B + (1+\beta) I$$

$$= \beta \frac{\partial I_B}{\partial I_C} + (1+\beta) \frac{\partial I_{CBO}}{\partial I_C}$$

$$\frac{\partial I_{CBO}}{\partial I_C} = \frac{1 - \beta\, \partial I_B / \partial I_C}{(1+\beta)}$$

$$\boxed{S = \frac{\partial I_C}{\partial I_{CBO}} = \frac{1+\beta}{1 - \beta\, \partial I_B / \partial I_C}}$$ **

↳ stability factor or sensitivity

Question-02

H... ...termine the transistor region of operation ?

(... ...stor)

...ctive region

...ply KVL in i/p loop &

...ermine I_B

... $I_E = (1+\beta)I_B$

... βI_B

③ apply KVL in output loop &

compute V_{CE}

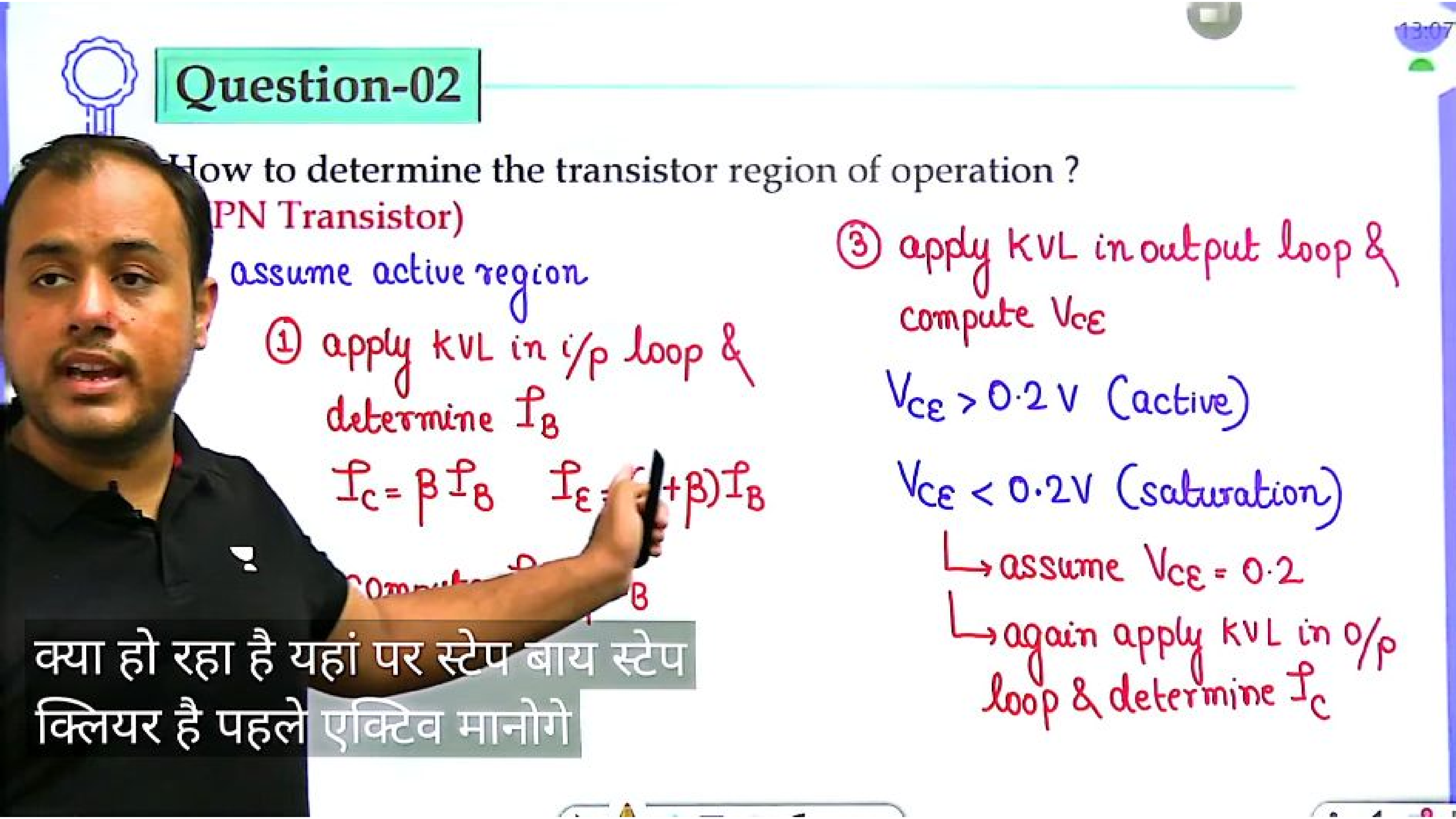
$V_{CE} > 0.2\,V$ (active)

$V_{CE} < 0.2V$ (saturation)

↳ assume $V_{CE} = 0.2$

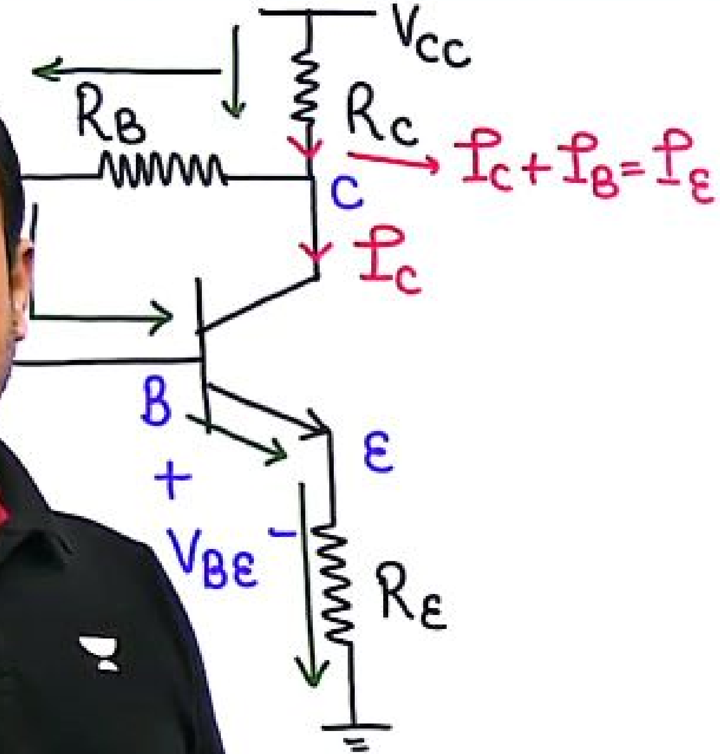
↳ again apply KVL in o/p loop & determine I_C

तो आपको दोबारा से इस निकलना पड़ेगा
ठीक है भाई आप लोगों को दोबारा से इस



13:07
Question-02
How to determine the transistor region of operation ?
PN Transistor)
assume active region
① apply KVL in i/p loop & determine I_B
I_C = β I_B
③ apply KVL in output loop & compute V_CE
V_CE > 0.2 V (active)
V_CE < 0.2V (saturation)
→ assume V_CE = 0.2
→ again apply KVL in o/p loop & determine I_C
क्या हो रहा है यहां पर स्टेप बाय स्टेप
क्लियर है पहले एक्टिव मानोगे

Collector to Base Bias Circuit

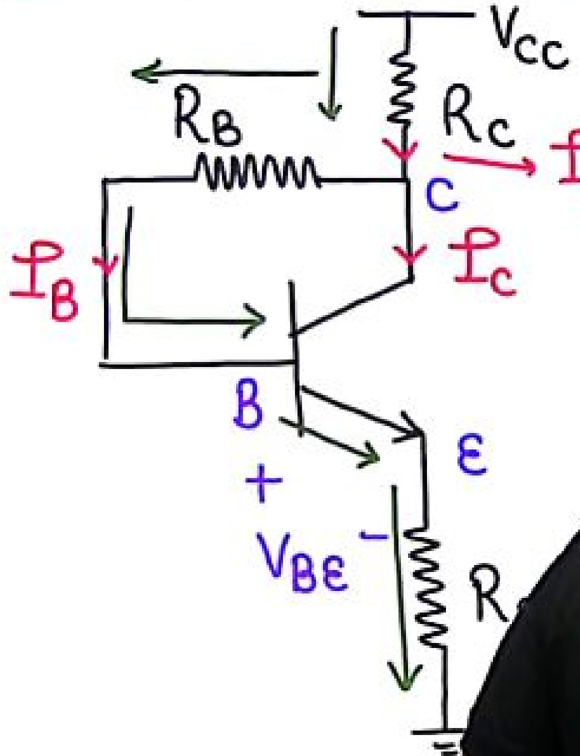


KVL

$$V_{CC} - (I_C + I_B)R_C - I_B R_B - V_{BE} - I_E R_E = 0$$

$$V_{CC} - (I_C + I_B)R_C - I_B R_B - V_{BE} - (I_C + I_B)R_E = 0$$

Collector to Base Bias Circuit



$I_C + I_B = I_E$

KVL

$V_{CC} - (I_C + I_B)R_C - I_B R_B$

$- I_E R_E = 0$

$(I_C + I_B)R_C - I_B R_B - V_{BE}$

$-(I_C + I_B)R_E = 0$

$0 - R_C - \frac{\partial I_B}{\partial I_C} R_C - \frac{\partial I_B}{\partial I_C} R_B - 0$

$- \frac{\partial I_B}{\partial I_C} R_E$

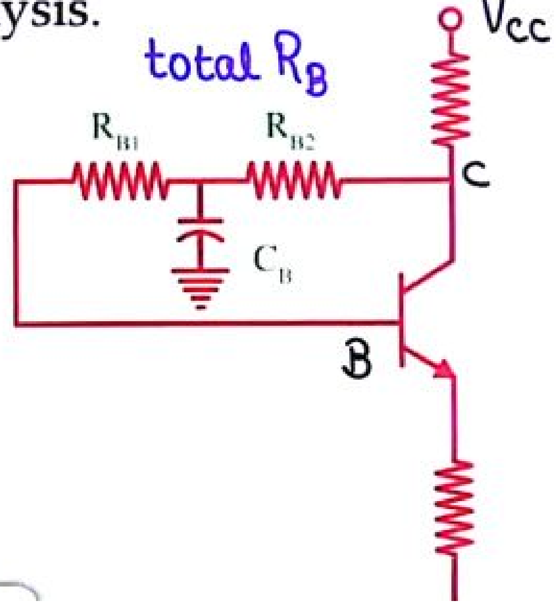
Drawback

Due to negative feed back from C to B through RB , VBE reduces due to which current and ut voltage also reduce. This leads to reduction in voltage gain.

To eliminate this capacitor is connected in the feedback path , which gets shorts during AC analysis.

Stability :

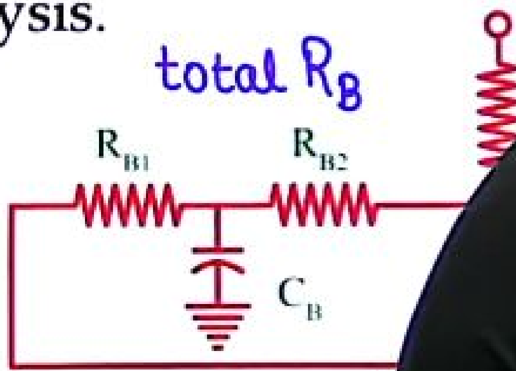
Self Bias > CB bia



Drawback

Due to negative feed back from C to B through RB , VBE reduces due to which current and output voltage also reduce. This leads to reduction in voltage gain.

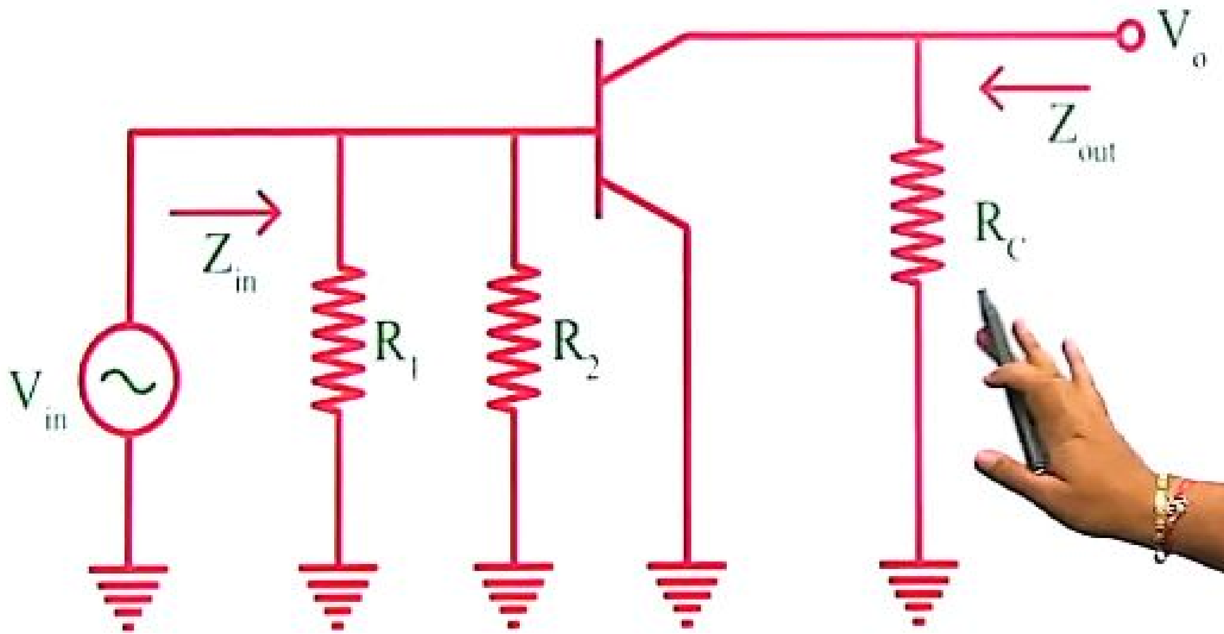
To eliminate this problem a capacitor is connected in the feedback pa which gets shorts circuited during AC analysis.



Stability :
Self Bias > CB bias > Fixed Bias

AC Analysis

- Short circuit DC voltage source. (grounded)
- Short circuit all capacitors



12:49 0 KB/s 48%

MOSFET Transconductance

$$g_m = \mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_T) \qquad I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_T)^2$$

$$\frac{g_m}{I_D} = \frac{2}{V_{GS} - V_T} \Rightarrow g_m = \frac{2 I_D}{V_{GS} - V_T}$$

So depending on which parameter is kept constant, the relation of these three parameters with

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MOSFET Transconductance and MOSFET Small Signal Model Explained

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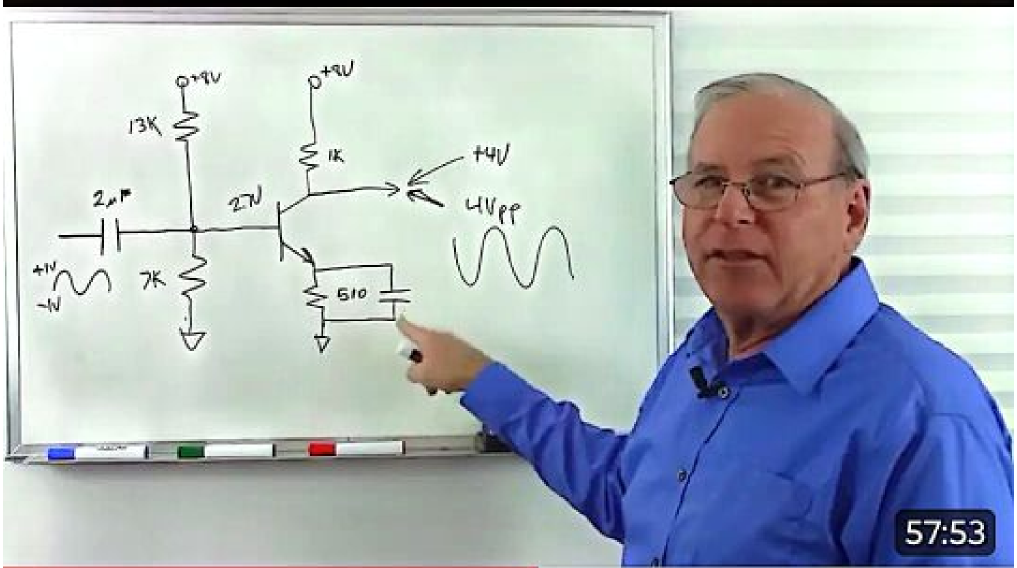
Thank you so much, just can't show enough gratitude through words. You're the real teacher.....☺🤗 Under...



Next:MOSFET Transconductance | Quiz # 309
Analog Electronics • 154/328

MOSFET Small Signal Model

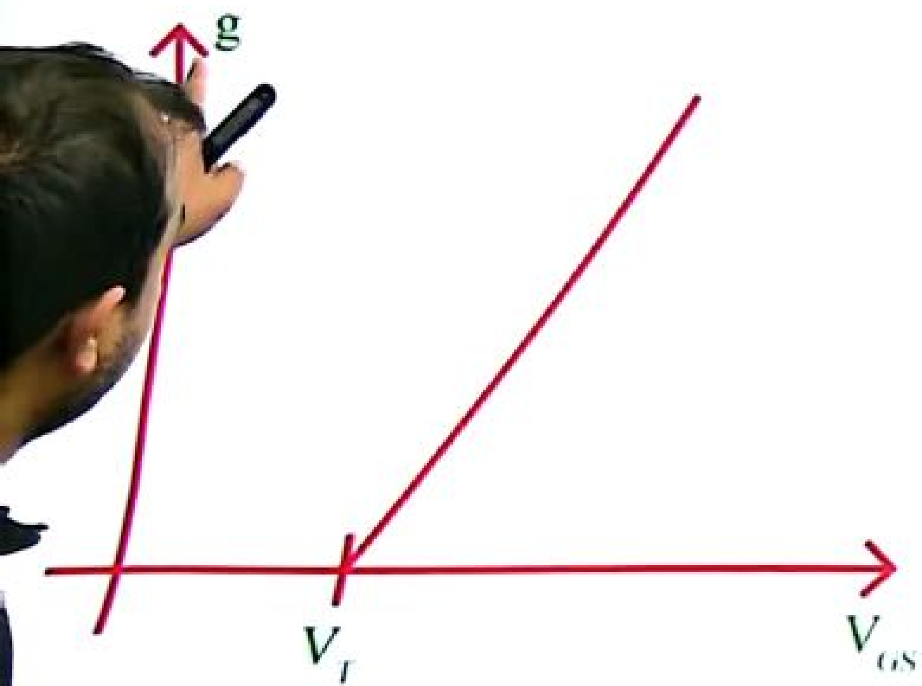
$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)\left[\underline{\underline{v_i}} + V_{DC} - V_T\right)^2$$

$v_i \ll V_{DC} - V_T$

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)\left[\underline{\underline{v_i^2}} + (V_{DC} - V_T)^2 + 2 v_i (V_{DC} - V_T)\right]$$

$$= \frac{1}{2}\underbrace{\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{DC} - V_T)^2}_{I_{DO}} + \underbrace{\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{DC} - V_T)}_{g_m}(v_i)$$

Channel Conductance



- conductance of region b/w drain & source

linear or triode region

$$I_D = \mu_n C_{ox} \frac{W}{L}\left[(V_{GS}-V_T)V_{DS} - \frac{V_{DS}^2}{2}\right]$$

V_{DS} = small, ignore $V_{DS}^2/2$

$$g_{ds} = \frac{\partial I_D}{\partial V_{DS}} = \mu_n C_{ox} W/L \,(V_{GS}-V_T)$$

$$r_{ds} = 1/g_{ds} = \frac{1}{\mu_n C_{ox} W/L\,(V_{GS}-V_T)}$$

13:04

MOSFET as an amp^r

saturation: $I_D = K_n (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$

$$g_{DS} = \frac{\partial I_D}{\partial V_{DS}} = K_n (V_{GS} - V_T)^2 \lambda$$

$$\boxed{\begin{aligned} r_d &= \frac{1}{K_n (V_{GS} - V_T)^2 \lambda} \\ r_d &\approx \frac{1}{\lambda I_D} \end{aligned}}$$

**

while

we ca $\lambda V_{DS})$

but this portant

to com

MOSFET as an amp^r

saturation: $I_D = K_n(V \ldots V_T)^2(1+\lambda V_{DS})$

$g_{DS} = \frac{\partial I_D}{\partial V_{DS}} = K_n \ldots \lambda$

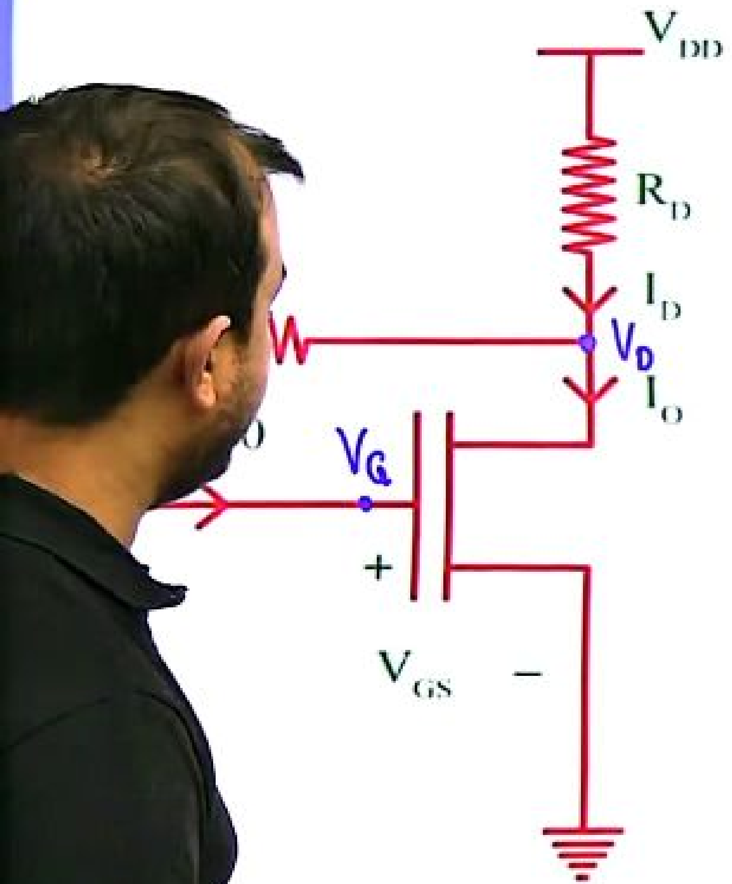
$$r_d = \frac{1}{K_n(V_{GS}-V \ldots}$$

$$r_d \approx \frac{1}{\lambda \ldots}$$

while computing I_D we can ignore $(1+\lambda V_{DS})$ but this term is important to compute r_o or r_d.

(2) Using Drain to gate feedback resistance

← Similar to C-B bias



$\because I_G = 0 \qquad V_{RG} = 0$

$\therefore V_D = V_G$

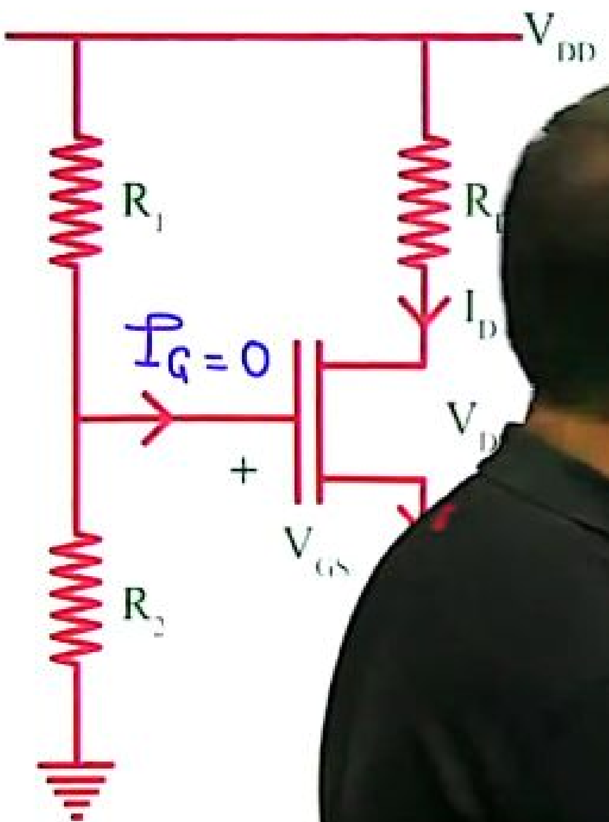
$V_D - V_S = V_G - V_S$

$V_{DS} = V_{GS}$

$\therefore V_{DS} > V_{GS} - V_T$ ← saturation region

→ When drain & gate terminal of a MOSFET are shorted, it operates in Saturation Region.

(3) Voltage Divider Biasing



R_1 & R_2 are in series

$$V_G = \frac{V_{DD} \times R_2}{R_1 + R_2}$$

$$V_S = I_D R_S$$

$$V_{GS} = V_G - V_S = (V_G - I_D R_S)$$

assume MOSFET operates in saturation region

$$I_D = K[V_{GS} - V_T]^2$$

$$I_D = K[V_G - I_D R_S - V_T]^2$$

solve quadratic & determine I_D

$$V_{GS} = V_G - I_D R_S$$

$> V_T$ (valid)

$< V_T$ (invalid)

12:03

apply KVL to determine

$$V_{DS} = V_{DD} - I_D(R_D + R_S)$$

$> V_{GS} - V_T$

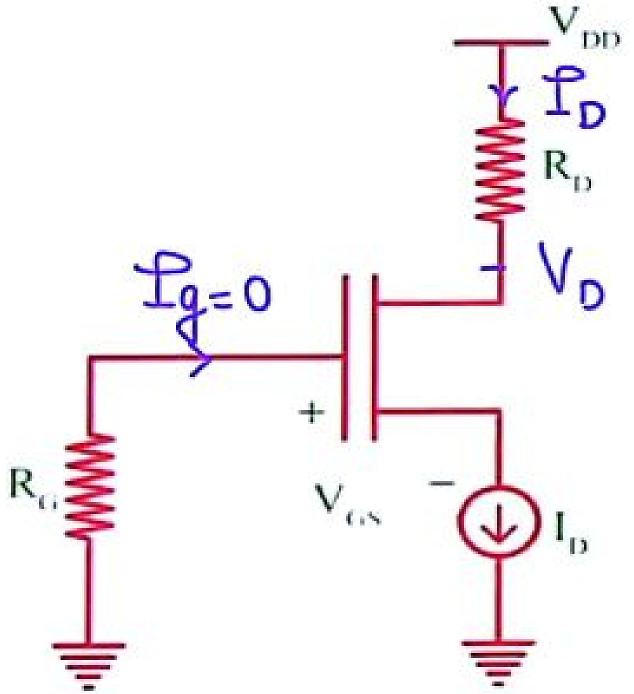
(sat)

$< V_{GS} - V_T$

(linear)

Solve for I_D again

(4) Constant Current Source



- we connect a current source at source terminal

  I_D = fixed

- $I_D = K[V_{GS} - V_T]^2$

  ↳ compute V_{GS}

  $V_G = 0$; $V_{GS} = 0 - V_S$

  $V_S = -V_{GS}$

- $V_D = V_{DD} - I_D R_S$
- $V_D = V_D - V_S$

Frequency Range	Coupling and Bypass Capacitor C_{C1}, C_{C2}, C_E [μF] (low f) high	Load and parasitic Capacitor C_d, C_T, C_L [pF] (high f) low
quency	**Consider** $X_C = 1/\omega C$ $\omega = low$ $C = high$ $X_C = finite$	**open** $X_C = 1/\omega C$ $\omega = low$ $C = low$ $X_C = high$
band	**short** $X_C = low$	**open** $X_C = high$
uency	**short** $\omega = high$ $C = high$ $X_C = low$	**Consider** $\omega = high$ $C = low$ $X_C = finite$

Q. Why does gain of amplifier reduce at low freq?

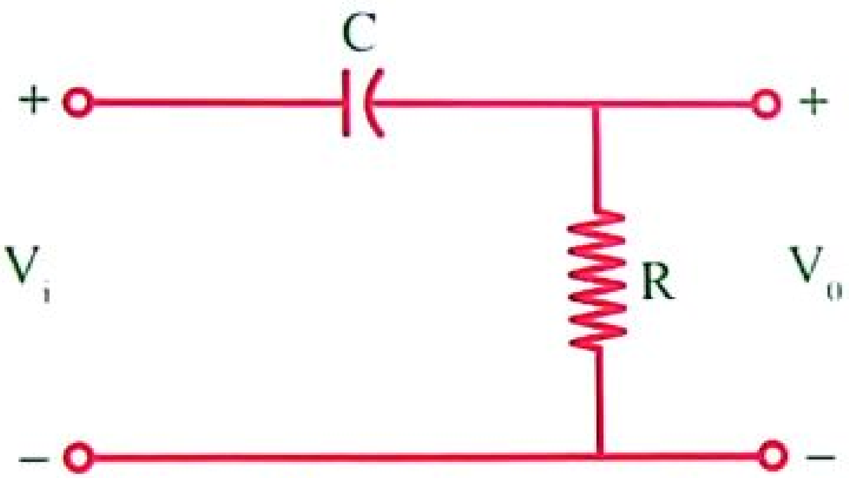
→ at low freq. we consider C_{c1}, C_{c2}, C_E

→ due to C_{c1} & C_{c2} in series there is voltage drop so $V_0 \downarrow$, gain $\downarrow$

→ C_E does not completely by-pass R_E $\therefore$ -ve f/b remains gain $\downarrow$

Low frequency response

LF response of a BJT is similar to a high pass filter



at low freq $X_C = 1/\omega C$ = high (open)

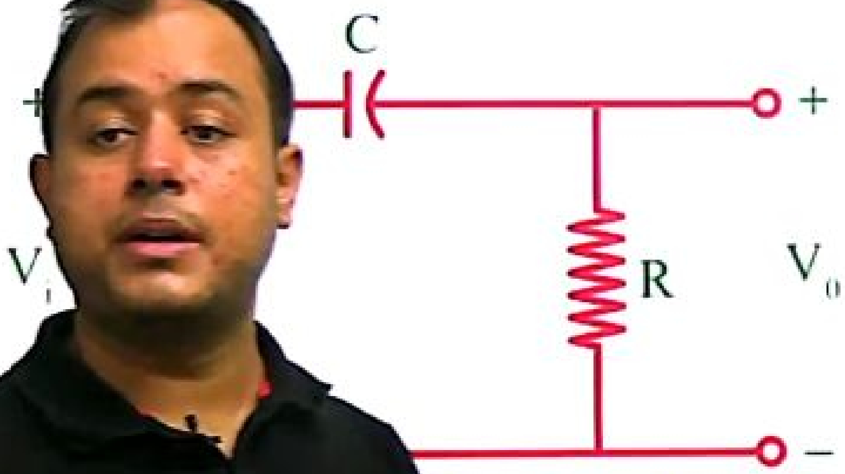
$i = 0 \quad V_0 = 0$

at high freq $X_C \to 0$ (short)

$V_0 = V_i$

Low frequency response

LF response of a BJT is similar to a high pass filter



$X_C = 1/\omega C$ = high (open)

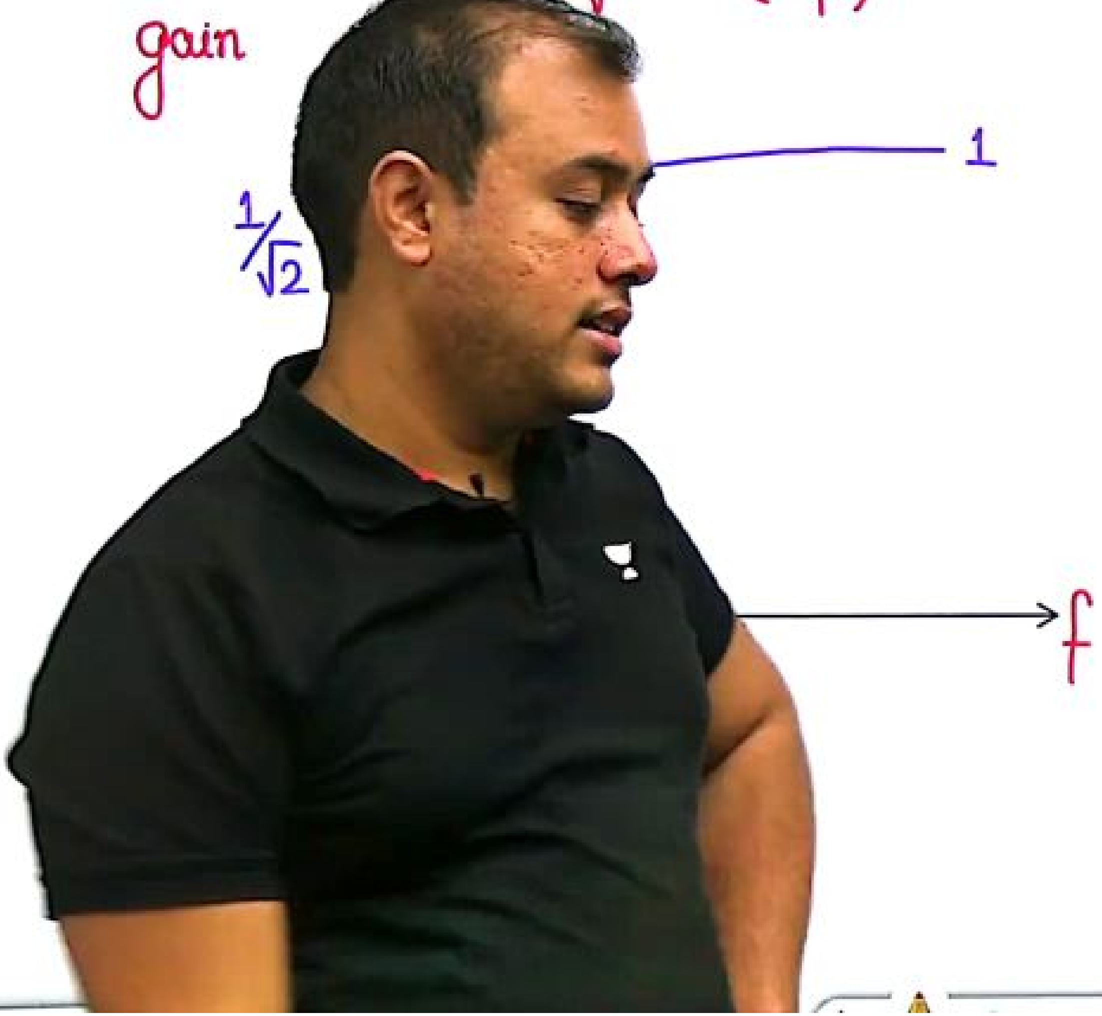
$X_C \to 0$ (short)

$$\frac{V_0}{V_i} = \frac{R}{R + 1/sC}$$

$$= \frac{sCR}{1 + sCR}$$

$$\frac{V_0}{V_i}(j\omega) = \frac{j\omega CR}{1 + j\omega CR}$$

$$\text{Gain (HPF)} = \frac{1}{\sqrt{1 + \left(\frac{f_L}{f}\right)^2}}$$

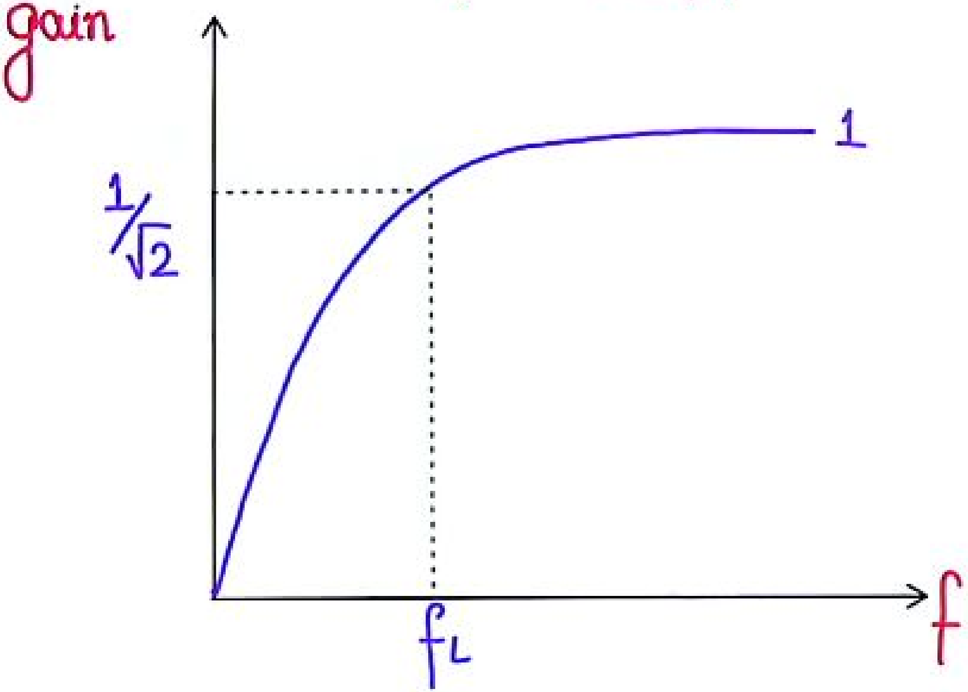


$$f_L = \frac{1}{2\pi RC}$$

R = equivalent resistance across cap. terminals

$$\text{Gain (HPF)} = \frac{1}{\sqrt{1 + \left(\frac{f_L}{f}\right)^2}}$$

$$f_L = \frac{1}{2\pi RC}$$



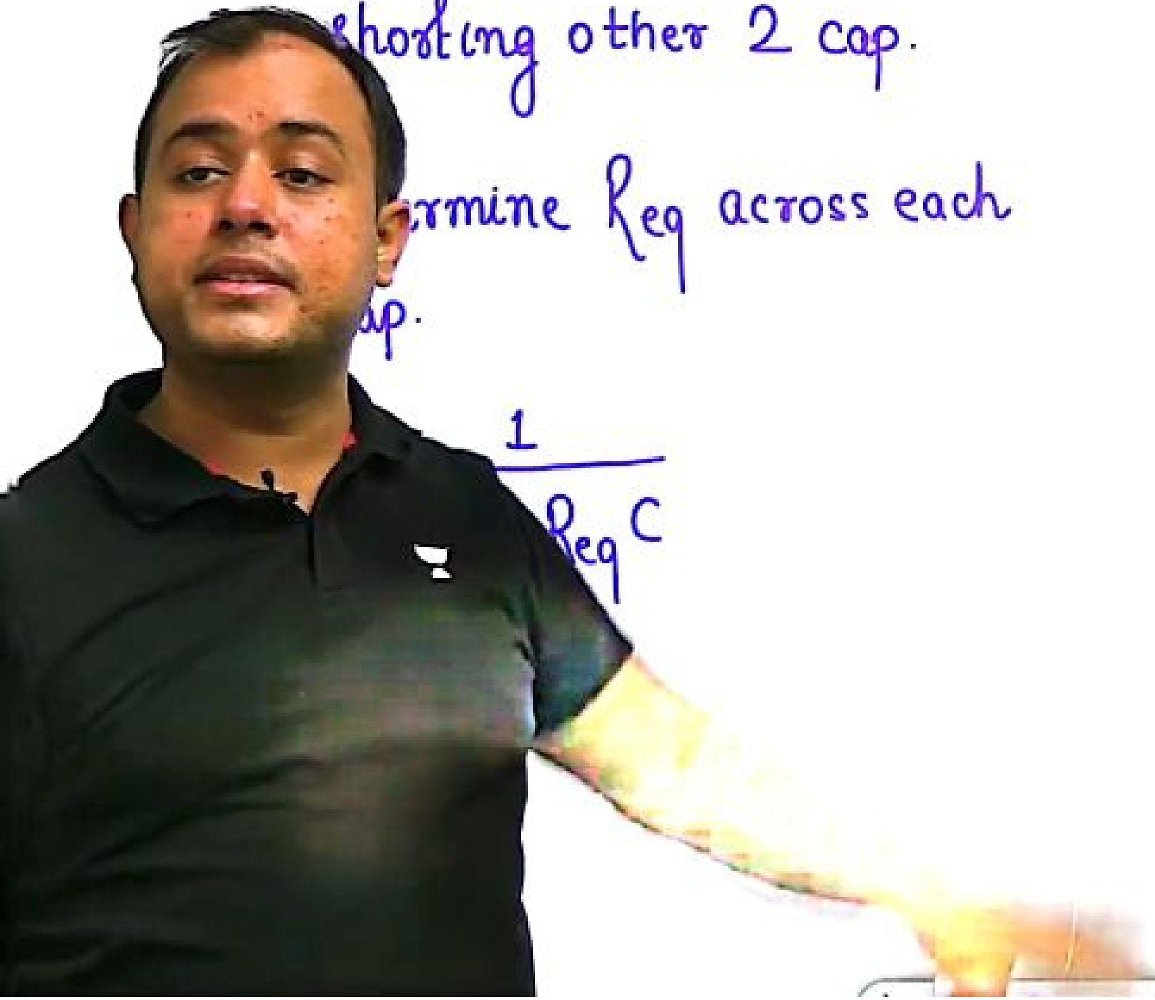
$R =$ … resistance

… terminals

12:24

- We have to consider one cap. at a time shorting other 2 cap.
- ...rmine R_{eq} across each ...ap.

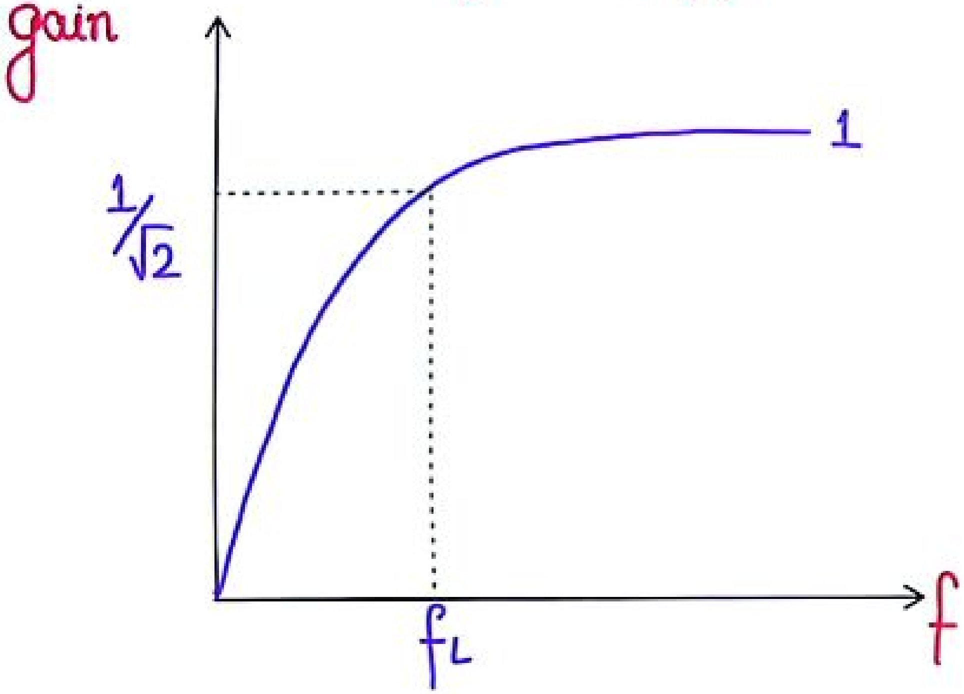
$$\frac{1}{R_{eq} C}$$



- we will get 3 values of f_L one corresponding to each cap.
- $f_L = \max[f_{L1}, f_{L2}, f_{L3}]$

12:23

$$\text{Gain (HPF)} = \frac{1}{\sqrt{1+\left(f_L/f\right)^2}}$$



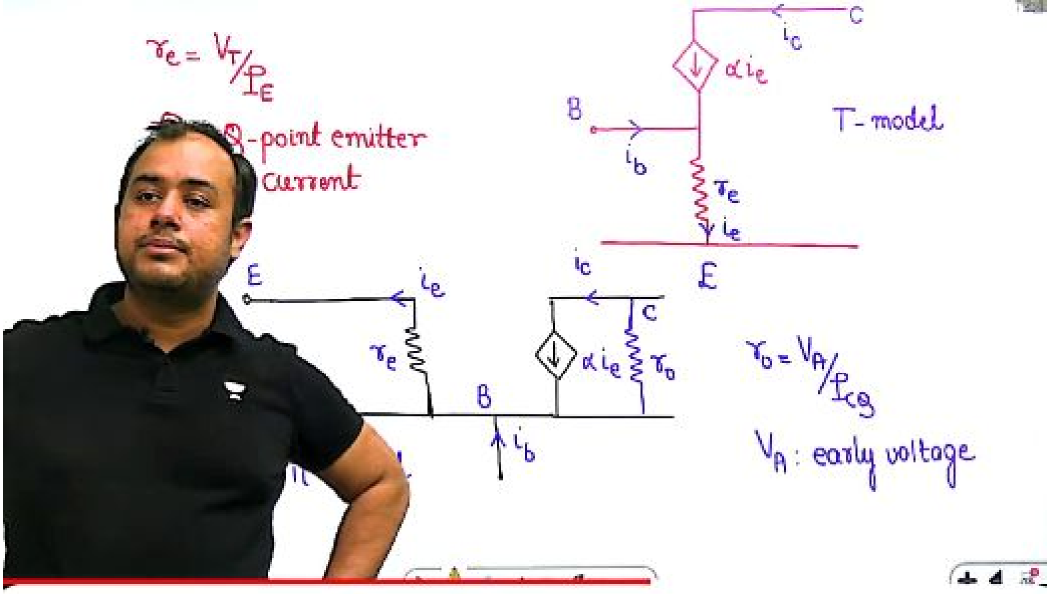
$$f_L = \frac{1}{2\pi RC}$$

R = equivalent r

across cap.

- We have to consider one cap. at a time shorting other 2 cap.
- determine R_{eq} across each cap.
- $f_L = \frac{1}{2\pi R_{eq} C}$

- we will get 3 values of f_L one corresponding to each ca
- $f_L = \max [f_{L1}, f_{L2}, f$



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Common Collector Configuration

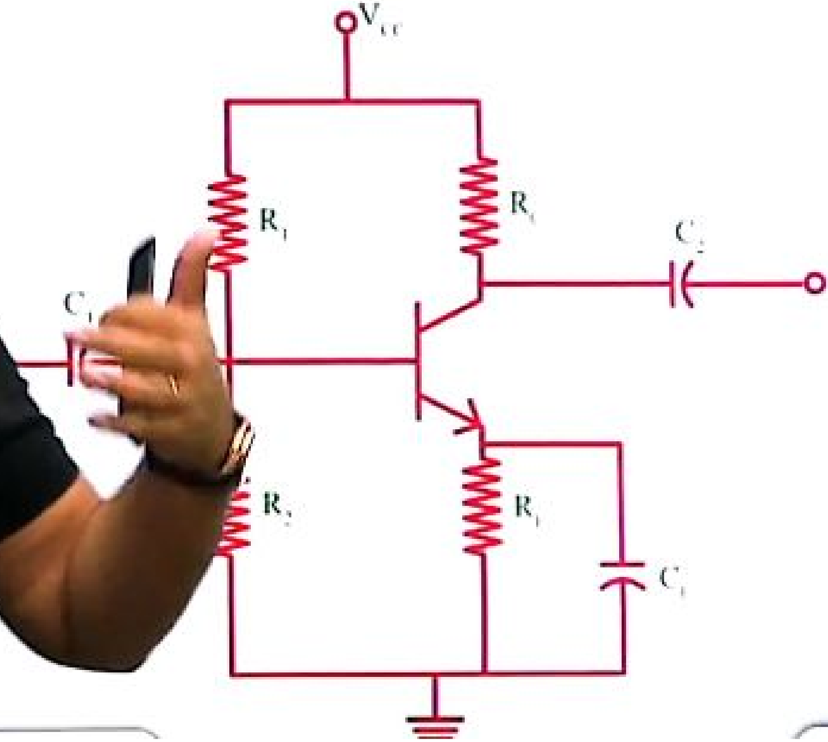
- The model is same as common emitt… …guration.
- The AC analysis is same for both N… …P transistor because input signal varies between positive… …ve value.

Voltage Divider Bias

Common Collector Configuration

- The model is same as common emitter configuration.
- The AC analysis is sa both NPN and PNP transistor because input signal varies ositive and negative value.

Voltage Divider



Common Collector Configuration

- The model is same as common emitter configuration.
- The AC analysis is same for both NPN and PNP transistor becau
input signal varies between positive and negative value.

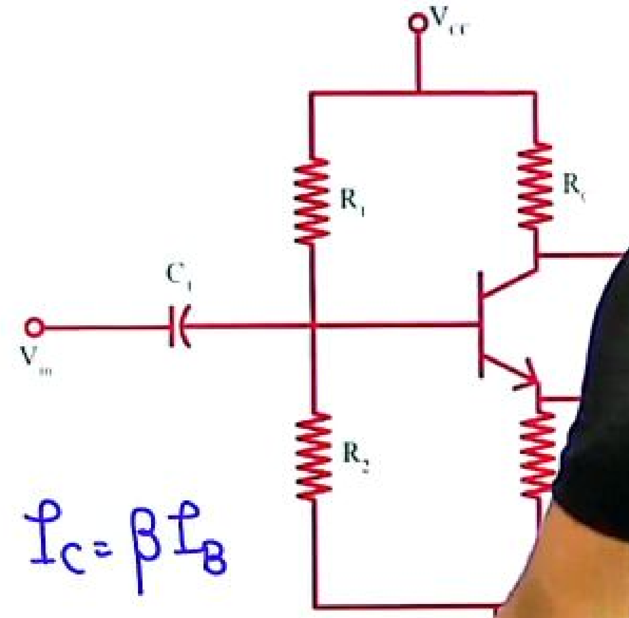
Voltage Divider Bias

dc analysis: C → open

$$V_{Th} = V_{CC}\frac{R_2}{R_1 + R_2}$$

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2}$$

KVL in BE loop: $I_B = ?$ $\quad I_C = \beta I_B$



Common Collector Configuration

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Voltage Divider Bias

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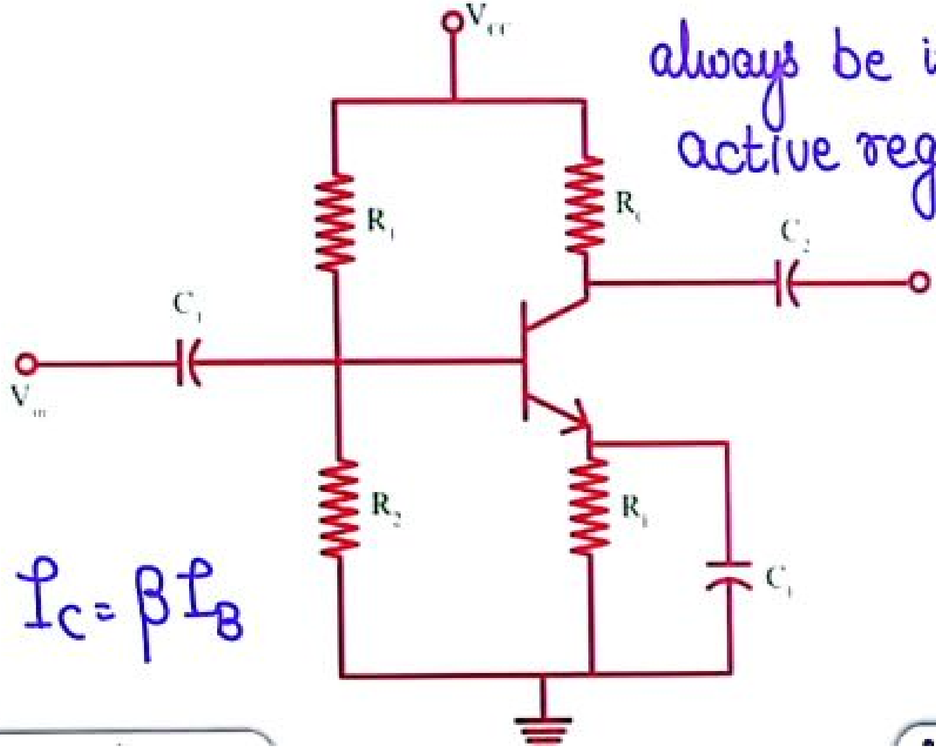
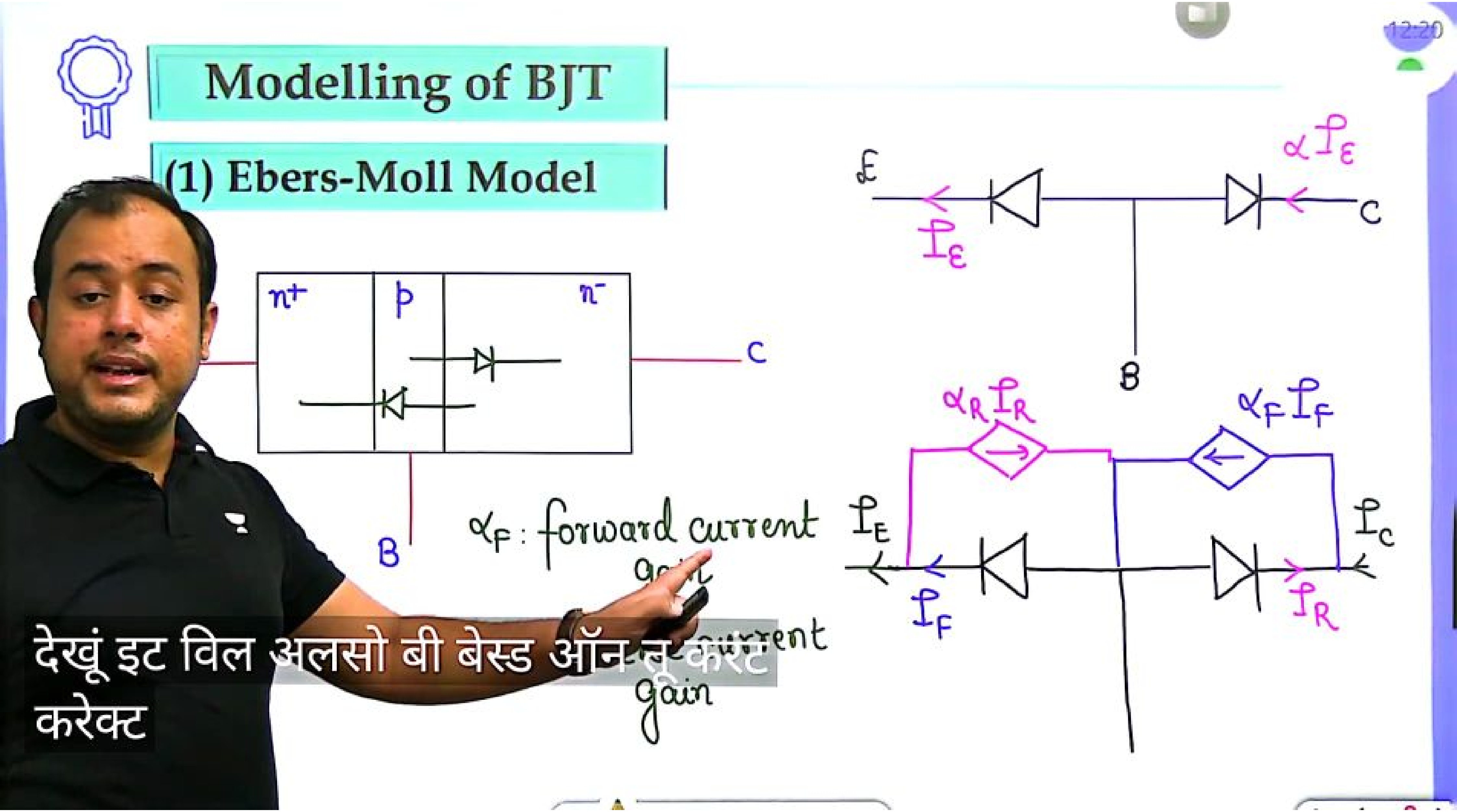
KVL in BE loop: $I_B = ?$ $I_C = \beta I_B$



always be in active region


Modelling of BJT
(1) Ebers-Moll Model
n+
p
n-
C
B
E
αI_E
I_E
C
B
α_R I_R
α_F I_F
I_E
I_C
I_F
I_R
α_F : forward current gain
current gain
देखूं इट विल अलसो बी बेस्ड ऑन टू करंट
करेक्ट
12:20